Mathematical Anatomy of Trading Systems

A SPY Multi-Leg Options Relative-Value System

Physical prediction, risk-neutral valuation and American exercise

A rigorous mathematical course

Volume II

September 2026

A route through this volume

This volume begins with the cash earned by a single option and a two-state replication calculation. It develops stochastic calculus, pricing probabilities, volatility coordinates, dynamic variance models and control as successive responses to difficulties in that simple trade.

Chapters 1–7 build contracts, replication and the distinction between physical and pricing probability. Chapters 8–11 develop surfaces and exposure geometry. Chapters 12–17 build volatility dynamics and the state information they need. Chapters 18–23 add American exercise, spread construction, execution, control, cash accounting and validation. The closing two-page sheet executes a declared specialization. The construction stays symbolic; no parameter calibration or empirical edge is asserted. A European laboratory is always distinguished from the actual American SPY contract.

How to read the mathematics

The prerequisites are rigorous linear algebra, multivariable calculus, basic analysis and elementary probability. Brownian integration, measure change, filtering and stopping are developed here. Work through the small replication and hedge examples before their general formulas. A price coordinate, a latent variance and a cash gain are different objects even when each is a scalar; their roles are explained where they first appear.

The main text introduces each object when its purpose becomes visible. Sections marked *Derivation* work through the algebra behind a result. Technical appendices retain domain restrictions, exceptional cases and longer proofs. A forward reference to an appendix is permission to defer that detail, not a prerequisite for understanding the next paragraph. The notation table is a reference; no symbol in it needs to be memorized first.

Contents

Principal notation **ix**

1 What an option trade earns **1**
1.1 Buying now and selling later . . . 1
1.2 Why premium received is not immediate profit . . . 1
1.3 Trading cash and marked equity . . . 1

2 Payoffs and the contracts that promise them **3**
2.1 The simplest payoff functions . . . 3
2.2 The information rule . . . 3
2.3 Information, underlying and contracts . . . 3

3 Why expected payoff is not automatically price **5**
3.1 A two-state market . . . 5
3.2 Solve the replication equations . . . 5
3.3 Two weights, two questions . . . 5

4 Replication and the Black–Scholes laboratory **7**
4.1 Why Brownian noise has a square-root time scale . . . 7
4.2 Conditional laws and Brownian integration . . . 7
4.3 Quadratic variation and Itô's formula . . . 8
4.4 A self-financing hedge . . . 8
4.5 Replication and the PDE . . . 8
4.6 What this laboratory is used for . . . 9

5 Physical probability: forecasting what happens **10**
5.1 Conditional averages under the physical law . . . 10
5.2 A physical stock model . . . 10
5.3 What must be forecast for an option trade? . . . 10

6 Pricing probability and discounted gains **12**
6.1 Return to the two-state weights . . . 12
6.2 No arbitrage and martingale measures . . . 12
6.3 From the pricing expectation to the formula . . . 13

7 Why P and Q solve different problems **14**
7.1 A change of probability as reweighting . . . 14
7.2 A density process . . . 14
7.3 Why the exponential density has a compensator . . . 14
7.4 Girsanov's drift change . . . 15
7.5 Risk premia and legitimate comparisons . . . 15

8 Implied volatility as a price coordinate **17**
8.1 Inverting a one-dimensional map . . . 17
8.2 Implied volatility is an inverse coordinate . . . 17

9 The volatility surface and its restrictions **19**
9.1 From a collection of quotes to a surface . . . 19
9.2 Static restrictions and what calendar consistency means . . . 19
9.3 European state-price density . . . 20
9.4 A controlled SSVI representation . . . 20

10 Greeks as derivatives of a pricing map **22**
10.1 Introduce the sensitivities one at a time . . . 22
10.2 Greeks are coordinate derivatives . . . 22
10.3 Taylor expansion and Itô P&L . . . 23

11 Portfolio exposure geometry **24**
11.1 Two options and a stock hedge . . . 24
11.2 From a line of solutions to a nullspace . . . 24
11.3 The exposure map and its nullspace . . . 24
11.4 Neutrality is local and model-dependent . . . 25

12 Why volatility becomes a state **27**
12.1 A scalar random variance before a diffusion model . . . 27
12.2 From random variance to stochastic volatility . . . 27
12.3 Why an unconstrained Gaussian variance is awkward . . . 27

13 Heston: a first dynamic volatility model **29**
13.1 Read the variance equation before solving it . . . 29
13.2 Heston dynamics and boundary conditions . . . 29
13.3 Two drivers and unspanned volatility . . . 30
13.4 A generator packages the Itô drift . . . 30
13.5 Correlation, characteristic transforms and pricing . . . 30

14 Physical forecasting and pricing calibration **32**
14.1 Observe a noisy proxy before introducing a belief . . . 32
14.2 Add unknown parameters, then alternative models . . . 32
14.3 The calibration problem at one date . . . 32

14.4 Calibration is a separate inverse problem . . . 33

15 Local volatility and the surface inverse problem 34
15.1 What local means . . . 34
15.2 Why a forward density equation appears . . . 34
15.3 Local volatility and Dupire's relation . . . 34

16 Why today's surface does not determine tomorrow's 36
16.1 The missing joint law: a finite example . . . 36
16.2 What a present option surface determines . . . 36
16.3 What state must be retained? . . . 36

17 Rough volatility and the information in a history 38
17.1 Why an exponential kernel has a short state description . . . 38
17.2 A kernel that cannot factor this way . . . 38
17.3 A Volterra Gaussian driver . . . 38
17.4 A representative rough Bergomi specification . . . 39
17.5 Conditional forward variance . . . 39
17.6 Physical forecasting is an additional specification . . . 40
17.7 Finite-dimensional approximation and its error . . . 40
17.8 Why short-maturity skew enters the comparison . . . 41
17.9 Model comparison tied to decisions . . . 41

18 American exercise as optimal stopping 43
18.1 Two exercise dates . . . 43
18.2 Stopping times and the Snell envelope . . . 43
18.3 Continuation and exercise regions . . . 44
18.4 Discrete dividends and early exercise . . . 44

19 Constructing spreads from forecasts and exposures 46
19.1 Assemble the information in a deliberate order . . . 46
19.2 A joint state and observation model . . . 46
19.3 The predictive law and the gain map . . . 47
19.4 What the forecast is claiming . . . 48
19.5 Choose a criterion before adding every constraint . . . 48
19.6 The objective and feasible decisions . . . 48
19.7 Existence, convex approximations and integer limits . . . 49
19.8 Payoff geometry produces names . . . 49

20 Execution, hedging, assignment and financing 51
20.1 Start from the cash account . . . 51
20.2 Multi-leg trading frictions . . . 51
20.3 The delta-hedged variance identity . . . 52
20.4 Discrete hedge error and transaction costs . . . 52

20.5 Exercise is a state transition . . . 53

21 Dynamic control of the option portfolio 55
21.1 Two trading dates before the general recursion . . . 55
21.2 The state and action spaces . . . 55
21.3 Bellman recursion and terminal delivery . . . 56
21.4 HJB, impulse control and stopping . . . 57
21.5 Receding horizon and stochastic actions . . . 57

22 Exact option P&L 59
22.1 Prove the ledger on one event at a time . . . 59
22.2 An event-ordered identity . . . 59
22.3 Premium, mark and delivery . . . 60
22.4 Greek attribution refines, but does not add to, option P&L . . . 60
22.5 Risk is a law of the full path . . . 61

23 Statistical validation of the options system 63
23.1 Fit, forecast and decision . . . 63
23.2 The claim to be investigated . . . 63
23.3 What is observed and what is tested . . . 64
23.4 Chronology and overlapping horizons . . . 64
23.5 A dependence-aware economic test . . . 64
23.6 Multiplicity and the right to stop believing . . . 65
23.7 Literature choices and maintained system . . . 65

A Probability and no-arbitrage qualifications 67
A.1 Full information and observable information . . . 67
A.2 Continuous-time no arbitrage . . . 67
A.3 Stopping integrability . . . 67
A.4 Conditional laws and integrability . . . 67

B Calculus, pricing and measure-change checks 69
B.1 The exponential density compensator . . . 69
B.2 Discounted gains . . . 69
B.3 Verification of the pricing PDE . . . 69

C Surface calculus and inverse conditioning 71
C.1 Technical note: SSVI admissibility restrictions . . . 71
C.2 Strike curvature in total-variance coordinates . . . 71
C.3 What a noisy inverse amplifies . . . 72

D Variance moments, roughness and memory 73
D.1 The square-root stationary density . . . 73
D.2 Fractional Brownian motion and regularity . . . 73

D.3 Short-maturity skew: a restricted comparison 74
D.4 The forward-variance curve is sufficient for the Gaussian driver 74

E Greek derivatives and constrained exposure **76**
E.1 The cancellation behind BSM Greeks 76
E.2 Weighted minimum-cost neutrality 76

F Stopping and ledger verification **78**
F.1 Why the obstacle has its signs 78
F.2 Delivery conservation 78
F.3 Terminal accounting and benchmark gain 78
F.4 One fully declared baseline within the model family 79

Complete SPY options trading system **83**

List of Figures

1.1 A gain depends on the whole holding rule. Later chapters fill in the market and contract events in the middle box. 2

2.1 The option lattice is a set of available contracts, not a continuous surface. Strike and maturity combinations create distinct exposure freedoms. 4

3.1 The price follows from reproducing the payoff. No estimate of the physical up probability is used. 6

5.1 A physical model is judged by future observations and policy gains. A present option-price fit answers a different question. 11

7.1 The pathwise covariance stays fixed. A change of measure is not permission to choose an unrelated diffusion. 15

9.1 Schematic total-variance slices. Drawing coordinates only set the geometry; they represent no calibration or empirical surface. 19

11.1 An affine Greek constraint set. The schematic shows one remaining degree of freedom; actual option space may be high-dimensional. 25

12.1 The new state is introduced because future option values can differ at the same stock price. 28

14.1 Observation and latent state. The posterior integrates quote noise, physical parameter uncertainty and pricing-model ambiguity. 33

14.2 Historical inference supplies a transition forecast. Quote calibration constrains a pricing map at a date. 33

16.1 The middle arrow cannot be recovered from a surface snapshot alone. 37

17.1 The curve retains horizon-dependent information that a single current variance cannot retain. 40

17.2 Model comparison concerns the future gain distribution needed by the trade. Present fit alone cannot select the physical dynamics. 42

18.1 For rough volatility, continuation depends on the enlarged state. For a short position, the holder's decision becomes an external assignment event. 45

19.1 Symbolic terminal payoff of a three-strike nullspace direction, shown per unit multiplier. The schematic is not the complete pathwise P&L of American legs. 50

20.1 The hedge loop includes assignment notices and physical delivery. A static Greek-neutral spread does not remain neutral automatically. 53

21.1 The generic state-action-transition loop. Assignment is part of the physical transition; only owned-option exercise is controlled. 56

22.1 Attribution has levels. Greek terms explain the option-mark branch and must not be added a second time to the ledger. 61

22.2 Nonlinear market risk, model risk and delivery funding risk enter the decision through distinct constraints or loss terms. 61

23.1 Information flow across an evaluation horizon. Outcomes become observations only after their realization. 64

23.2 Physical forecasts and risk-neutral values are different maps. A future pricing state must itself be forecast under $\mathbb{P}$. 66

Principal notation

Symbol	Meaning and space
$\mathbb{P}, \mathbb{Q}, \mathcal{G}_t$	Physical measure, selected equivalent pricing measure, observable filtration.
$S_t > 0, D_t, B_t$	SPY ex-dividend price, cumulative cash dividends per share, money-market account.
r_t, δ_t	Short rate and continuous dividend yield in auxiliary smooth models.
(K_j, T_j, χ_j, m_j)	Strike, expiry, call/put sign and positive contract multiplier.
c_j^A, c_j^E, C_j	Per-share American and European values; contract mark $C_j = m_j c_j^A$.
$q \in \mathbb{Z}^m, h \in \mathbb{R}, X, W$	Option contracts, SPY shares, cash balance, marked equity.
v_t, ζ_t, b_t	Latent variance, surface coordinates, posterior belief.
ϑ^P, ϑ^Q	Physical transition and pricing-law parameters.
$Z_t^{\mathbb{Q}}$	Radon–Nikodym density process for the measure change.
$k, \tau, w(k, \tau)$	Log forward moneyness, remaining maturity, total implied variance.
$H, \mathcal{K}_H$	Hurst parameter and Volterra kernel.
$\xi_t(u)$	Conditional forward variance at future time u, under a stated measure.
$G \in \mathbb{R}^{p \times m}$	Greek exposure matrix in chosen risk coordinates.
a, b, e, A	Filled option trade, filled stock trade, signed exercise removal, assignment count.
$\mathcal{V}_j^M(t, z)$	American pricing functional in model M at state z.
$\Pi_{t,t+h}^{q,\pi}$	Full financed gain of the option portfolio with hedge/exercise policy π.
$\mathcal{M}, \mathcal{U}$	Model family and remaining statistical ambiguity set.
$V_t(s), \pi_t^\star$	Control value and criterion-optimal feasible policy.

Chapter 1

What an option trade earns

What problem are we solving?

An option payoff diagram describes what a contract can deliver, but an investor pays a premium and may close or hedge the position before expiry. We begin with a cash comparison that separates the price paid from the later gain.

1.1 Buying now and selling later

Let C_t be the cash price of one option contract today. If it remains alive and is sold at $u > t$ for C_u, the unfinanced gain is $C_u - C_t$, before transaction costs. Let $R_{t,u} > 0$ be the gross growth factor of the cash benchmark over the interval. The financed gain is

$$\Pi_{t,u} = C_u - R_{t,u}C_t - K_{t,u}.$$

The term $K_{t,u}$ is the total cost expressed in date-u cash units. The initial premium could otherwise have earned $R_{t,u}C_t$, which explains the subtraction. If the contract is short, the mark exposure reverses sign, but the actual costs and financing still have to be included.

This formula is deliberately a small laboratory: one contract, no early exercise and no changing hedge. The complete system will replace it by an event-by-event ledger. Its lesson survives every refinement: a future price or payoff is not itself the profit.

1.2 Why premium received is not immediate profit

Selling one option credits cash C_t while creating a liability marked $-C_t$. Buying it debits cash while creating an equal asset. The account's equity changes at the trade only by costs. We now express this cancellation for multiple contracts and a stock hedge, using a vector only to sum the cash.

1.3 Trading cash and marked equity

Let X_t denote cash, h_t SPY shares and $q_t \in \mathbb{Z}^m$ option contracts. Marked equity is

$$W_t = X_t + h_t S_t + q_t^\top C_t. \tag{1.1}$$

Posted collateral is a partition of assets or cash, not a loss by itself. Reserve and margin maps will constrain available financing.

At fixed quotes, filled option changes $a_t \in \mathbb{Z}^m$, stock changes $b_t \in \mathbb{R}$, and cost $K_t^{\mathrm{tr}}(a_t, b_t) \geq 0$ produce

$$q_t^+ = q_t^- + a_t, \quad h_t^+ = h_t^- + b_t, \quad X_t^+ = X_t^- - a_t^\top C_t - b_t S_t - K_t^{\mathrm{tr}}. \tag{1.2}$$

Thus $W_t^+ - W_t^- = -K_t^{\mathrm{tr}}$. Receiving a short-option premium raises cash and creates an equal marked liability; it does not create profit.

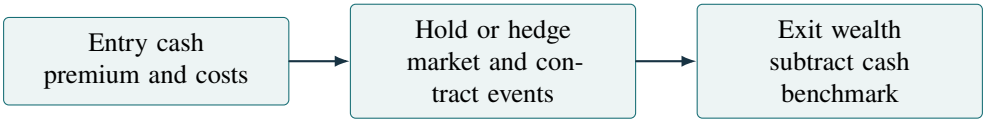


Figure 1.1: A gain depends on the whole holding rule. Later chapters fill in the market and contract events in the middle box.

What we gained

- A financed gain for a single option held without exercise.
- Marked equity containing cash, stock and signed option positions.
- A trade identity that distinguishes premium from profit.

Where this goes next

To generalize the holding rule, we need the contract itself: strike, expiry, payoff, multiplier and the right to exercise.

Chapter 2

Payoffs and the contracts that promise them

What problem are we solving?

The symbol for an option price hides legal and cash-flow terms. We describe those terms mathematically before asking what a fair price or a useful forecast should be.

2.1 The simplest payoff functions

A call permits buying a share at strike $K > 0$. If exercised when the share price is s, its immediate economic value is $(s - K)_+$, where $x_+ = \max(x, 0)$. A put permits selling at K, with value $(K - s)_+$. The kink at $s = K$ is the first reason option risk differs from a linear stock holding: sensitivity changes with the market state.

A European contract permits exercise only at its expiry T. An American contract permits exercise on its allowed dates up to T. A cash-settled contract pays cash; a physically settled contract exchanges shares and strike cash. Equal immediate values do not make their intermediate obligations equal. For now we record these distinctions; Chapter 18 will determine when exercise is valuable, and Chapter 20 will write its cash map.

2.2 The information rule

Write $\mathcal{G}_t$ for everything released by date t: quotes, dividend announcements and received account notices. A decision at t can depend on that information but not on tomorrow's prices or an assignment notice that has not arrived. Expectations conditional on $\mathcal{G}_t$ average only the remaining uncertainty. Formal probability definitions are collected in Appendix A.

2.3 Information, underlying and contracts

Let $S_t > 0$ be the ex-dividend SPY share price and D_t the cumulative cash dividend per share. The gain from one share is $\mathrm{d}S_t + \mathrm{d}D_t$. Let $B_t = \exp(\int_0^t r_u \,\mathrm{d}u) > 0$ be a money-market account, where r_t is a locally integrable short rate. Actual borrowing and lending may have different rates; this ideal account is the baseline for pricing and a benchmark for financed gains.

Contract $j \in \{1, \ldots, m\}$ is indexed by strike $K_j > 0$, expiry T_j, type $\chi_j = 1$ for a call and $\chi_j = -1$

for a put, and multiplier $m_j > 0$. Its per-share exercise value is $g_j(s) = [\chi_j(s - K_j)]_+$. Its market bid and ask are $c_{j,t}^{\text{bid}} \le c_{j,t}^{\text{ask}}$, and a declared reference mark $c_{j,t}$ lies within that interval. Contract mark is $C_{j,t} = m_j c_{j,t}$. A signed contract position q_j has marked value $q_j C_j$.

The universe comprises standard listed SPY ETF options, which are American-style and settle by delivery of shares [5, 15]. It excludes specially negotiated alternatives and adjusted deliverables unless their terms are explicitly included. The multiplier remains symbolic; adjusted contracts require their actual delivery basket in place of m_j shares. A European cash-settled index option is a different security.

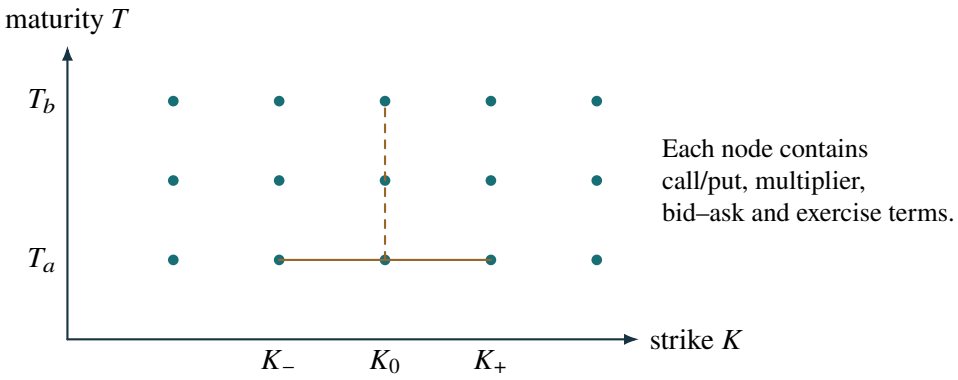


Figure 2.1: The option lattice is a set of available contracts, not a continuous surface. Strike and maturity combinations create distinct exposure freedoms.

What we gained

- Call and put payoffs and their nonlinear dependence on stock price.
- The differences between exercise style and settlement method.
- An identified SPY option universe, quoted marks and information timing.

Where this goes next

The payoff is now defined, so we can take its physical expectation. That average is useful for forecasting, but it does not yet explain the contract's price.

Chapter 3

Why expected payoff is not automatically price

What problem are we solving?

A physical expectation weights future payoffs by their likelihood. A market price also reflects the cost of reproducing those payoffs and how different states are valued. A one-period two-state market makes the distinction visible without stochastic calculus.

3.1 A two-state market

Suppose a non-dividend-paying stock costs $S_0 > 0$. At the next date it will be uS_0 or dS_0, where $0 < d < R < u$, and cash grows by a factor $R > 0$. Let the physical up probability be $p \in (0, 1)$. A claim pays H_u in the up state and H_d in the down state. Its physical expected payoff is $pH_u + (1-p)H_d$. The inequality $d < R < u$ makes the cash payoff lie strictly between the two stock returns; otherwise a simple stock–cash trade would produce an arbitrage.

3.2 Solve the replication equations

Hold Δ shares and put b cash units into the account. Matching both future claim payoffs requires

$$\Delta u S_0 + bR = H_u, \qquad \Delta d S_0 + bR = H_d.$$

Subtracting eliminates cash and gives $\Delta = (H_u - H_d)/(S_0(u - d))$. Substitute back to obtain $b = (uH_d - dH_u)/(R(u - d))$. Therefore the cost of reproducing the claim is

$$C_0 = \Delta S_0 + b = \frac{1}{R}\{p^* H_u + (1 - p^*)H_d\}, \qquad p^* = \frac{R - d}{u - d}. \tag{3.1}$$

If a traded claim with the same obligations had a different frictionless price, buy the cheaper side and sell the more expensive side. Their future cash flows cancel in both states, leaving the initial price difference. No arbitrage therefore forces the replication cost as the price.

3.3 Two weights, two questions

The weight p^* was derived from traded stock and cash, not from the frequency p of stock rises. In fact

$$\mathbb{E}^{\mathbb{P}}[H] - RC_0 = (p - p^*)(H_u - H_d).$$

Here $\mathbb{P}$ denotes physical probabilities. A nonzero expected financed gain is compatible with the unique no-arbitrage price because the claim still has risk. This calculation is the elementary origin of the distinction between forecasting and pricing probabilities.

Key idea

Replication determines a price by matching every payoff state. Physical expectation determines an average by weighting possible outcomes. Neither operation can silently replace the other.

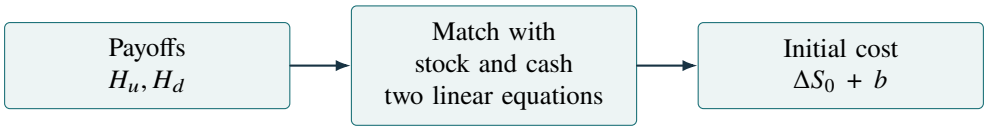


Figure 3.1: The price follows from reproducing the payoff. No estimate of the physical up probability is used.

What we gained

- A complete two-state replication calculation.
- A pricing weight derived from stock and cash prices.
- An explicit example of a physical expected gain at a no-arbitrage price.

Where this goes next

Continuous trading is a limit in which many small stock moves replace the two branches. We develop the required stochastic calculus before deriving the Black–Scholes replication equation.

Chapter 4

Replication and the Black–Scholes laboratory

What problem are we solving?

A hedge must now adjust as the stock moves continuously. Ordinary calculus misses the accumulated effect of squared random increments; we derive that correction before using it to cancel the option's stock risk.

4.1 Why Brownian noise has a square-root time scale

Over a short interval of length Δt, a Brownian increment has mean zero and variance Δt. Its size is therefore of order $\sqrt{\Delta t}$. Squaring produces a quantity of order Δt, which can accumulate to a nonzero limit across many intervals. This is why a second derivative survives in the continuous-time chain rule.

A stochastic differential equation states how drift accumulates over ordinary time and noise accumulates through these increments. It means an integral equation; it does not assert that a Brownian path has an ordinary derivative.

4.2 Conditional laws and Brownian integration

A martingale formalizes a conditionally fair fluctuation. Brownian stochastic integrals provide such fluctuations when the integrand is selected before each increment and has the required integrability.

Conditional expectation $\mathbb{E}^{\mathbb{P}}[Y \mid \mathcal{G}_t]$ averages unknown outcomes while holding present information fixed. It is defined by measurability and $\mathbb{E}^{\mathbb{P}}[\mathbf{1}_A \mathbb{E}^{\mathbb{P}}(Y \mid \mathcal{G}_t)] = \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A Y]$ for $A \in \mathcal{G}_t$. A martingale M_t is integrable, adapted, and satisfies $\mathbb{E}[M_u \mid \mathcal{F}_t] = M_t$ for $u \geq t$. A local martingale obeys this property after a sequence of stopping times tending to infinity; it need not be a true martingale.

Brownian motion B_t^W has continuous paths, starts at zero, and has independent Gaussian increments with variance equal to elapsed time. We use W_t^S, W_t^v for Brownian drivers later, reserving unadorned W_t for wealth. For a predictable square-integrable process H_t, the Itô integral is the mean-square limit of left-endpoint sums. Independence of Brownian increments gives the isometry

$$\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}B_t^W\right)^2\right] = \mathbb{E}\int_0^T H_t^2 \, \mathrm{d}t. \tag{4.1}$$

This identity ensures a well-defined integral and a centered martingale for square-integrable integrands. Local integrability gives a local martingale. Left endpoints encode that a position is selected before its next price shock.

4.3 Quadratic variation and Itô's formula

Start with an ordinary Taylor expansion of a smooth function over one small increment. Sum those expansions along a partition. The first-order terms become ordinary and stochastic integrals; the squared Brownian terms converge to elapsed time. The formula below is the resulting chain rule.

On a partition with mesh tending to zero, Brownian squared increments sum in probability to elapsed time: $\sum_j (\Delta B_j^W)^2 \to T$. The expectation of the sum is T; its variance is $2\sum_j (\Delta t_j)^2 \to 0$, proving the statement for deterministic partitions. Ordinary differentiability would instead make the sum tend to zero.

The regularity notation $C^{1,2}$ means one continuous time derivative and two continuous spatial derivatives. Those are the derivatives used in this chain rule. For $dX_t = b_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}B_t^W$, a second-order Taylor expansion of a $C^{1,2}$ map $f(t,x)$ contains a surviving term $\frac{1}{2}f_{xx}(\Delta X)^2$. Terms with $\Delta t\Delta B$ and $(\Delta t)^2$ vanish under localization and regularity, while $(\Delta X)^2$ sums to $\int \sigma^2\,\mathrm{d}t$. Therefore

$$\mathrm{d}f(t,X_t) = \left(f_t + b_t f_x + \frac{1}{2}\sigma_t^2 f_{xx}\right)\mathrm{d}t + \sigma_t f_x\,\mathrm{d}B_t^W. \tag{4.2}$$

For a vector X with diffusion matrix A_t, replace the second-order term by $\frac{1}{2}\operatorname{tr}(A_t A_t^\top \nabla^2 f)$. The derivatives must exist on the relevant region and integrals must be locally finite. At a dividend or exercise jump, add the exact jump $f(t,X_t) - f(t,X_{t-})$; it is not described by the continuous diffusion formula.

4.4 A self-financing hedge

The two-state hedge matched the coefficient of the up–down payoff difference. In continuous time we instead match the coefficient of the Brownian shock. The delta Δ_t is simply the number of shares needed for that local match. Cash finances the difference between the option value and those shares. Rebalancing must use the portfolio's own cash; inserting outside funds would not establish a replication price.

4.5 Replication and the PDE

Temporarily assume a frictionless European claim with payoff $g(S_T)$, continuous trading, no jumps, constant volatility $\sigma > 0$, deterministic rate r and continuous dividend yield δ. Under $\mathbb{P}$,

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^P, \qquad \mathrm{d}D_t = \delta S_t\,\mathrm{d}t. \tag{4.3}$$

Here μ is the ex-dividend physical price drift. Seek a $C^{1,2}$ value $c(t,S_t)$ replicated by Δ_t shares and cash $c - \Delta S$. Self-financing replication requires

$$\mathrm{d}c = \Delta(\,\mathrm{d}S + \delta S\,\mathrm{d}t) + r(c - \Delta S)\,\mathrm{d}t. \tag{4.4}$$

Derivation: match the random and deterministic terms

Itô's formula gives diffusion coefficient $c_s \sigma S$, so $\Delta = c_s$. Matching drifts cancels $\mu S c_s$ and gives

$$c_t + (r - \delta) s c_s + \frac{1}{2}\sigma^2 s^2 c_{ss} - rc = 0, \qquad c(T, s) = g(s). \tag{4.5}$$

The cancellation explains why physical drift does not enter this complete-market replication price. It does not imply that physical drift is irrelevant to the P&L of an unhedged option.

The derivation assumes the required rebalancing is feasible at zero cost and that the resulting discounted hedging gains are true martingales. With costs, discrete rebalancing or stochastic unspanned volatility, the replication argument no longer delivers a unique costless hedge.

Read the pricing equation term by term

In (4.5), c_t is change from passage of calendar time; $(r-\delta)sc_s$ is the stock-related drift after financing and dividends; $\sigma^2 s^2 c_{ss}/2$ is the curvature contribution from squared shocks; and $-rc$ is financing of the claim's value. The terminal condition fixes the contract being replicated. The physical stock drift cancels because the hedge matches its stock exposure, not because expected stock returns are zero.

4.6 What this laboratory is used for

It supplies a monotone volatility coordinate, a differentiable sensitivity benchmark, and a transparent decomposition of hedged gamma versus realized variance. It is not the direct mark of a standard SPY American contract when early exercise matters. Discrete cash dividends replace continuous yield by jump conditions, and American exercise replaces the terminal-value PDE by an obstacle problem in Chapter 18.

What we gained

- Brownian integration and the quadratic-variation correction.
- A self-financing delta hedge derived by matching shocks.
- The Black–Scholes PDE and the assumptions behind its price.

Where this goes next

Replication removed physical drift from this price. A trader still needs physical probabilities to forecast an unhedged or imperfectly hedged gain; we now develop that law explicitly.

Chapter 5

Physical probability: forecasting what happens

What problem are we solving?

The pricing equation describes the cost of replication, not which market paths will occur. To evaluate a trade, we must forecast paths under a law for actual outcomes.

5.1 Conditional averages under the physical law

The symbol $\mathbb{P}$ denotes the physical probability model. For possible future states $z_1, \ldots, z_m$ and probabilities p_i conditional on today's information, a gain function g has mean $\sum_i p_i g(z_i)$. In a continuous state we replace the sum by an integral. Conditioning says which information has already been used; it is not a second price adjustment.

5.2 A physical stock model

In the simplest diffusion, $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W^P$. Here μ is the ex-dividend expected rate of stock-price change, σ is volatility, and W^P is Brownian under $\mathbb{P}$. With constant coefficients and $S_t = s$, applying the chain rule to $\log S$ gives

$$S_{t+h} = s \exp\{(\mu - \sigma^2/2)h + \sigma\sqrt{h}\,Z\}, \qquad Z \sim \mathcal{N}(0,1).$$

The subtraction of $\sigma^2/2$ is the Itô correction for the second derivative of the logarithm. Since $\mathbb{E}e^{aZ} = e^{a^2/2}$, the conditional mean is $se^{\mu h}$. This law can forecast a stock-linked payoff, but substituting μ for $r - \delta$ in a replication price changes its meaning.

5.3 What must be forecast for an option trade?

For expiry-only cash proceeds, a terminal stock distribution may suffice. For sale before expiry, the future option mark also depends on the pricing conditions then prevailing. A hedge uses intermediate stock moves. An American contract may exercise before the horizon. Thus a realistic gain often requires a joint path law, not just a forecast of S_T. We will add variance, surface and observation states only when each becomes necessary. In particular an instantaneous variance v_t will mean the

coefficient of $\mathrm{d}t$ in the quadratic variation of log stock price; in the constant-volatility laboratory it is simply σ^2.

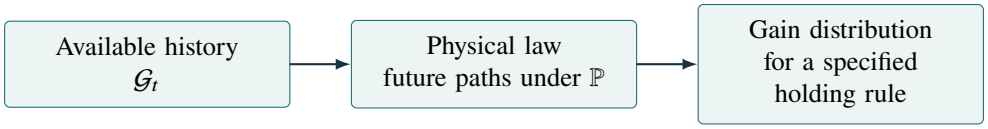


Figure 5.1: A physical model is judged by future observations and policy gains. A present option-price fit answers a different question.

What we gained

- Physical conditional expectations distinguished from prices.
- The physical lognormal stock transition derived with Itô calculus.
- A reason to forecast paths and future pricing conditions.

Where this goes next

The pricing weights in the two-state example were also probabilities. We now generalize them to a pricing law and obtain the Black–Scholes formula from it.

Chapter 6

Pricing probability and discounted gains

What problem are we solving?

Solving a replication PDE is one route to a price. A second route averages discounted payoffs under probabilities that make traded discounted gains fair. We first identify those probabilities, then connect the expectation to the same PDE.

6.1 Return to the two-state weights

The weights $p^*, 1 - p^*$ from (3.1) satisfy $p^*u + (1 - p^*)d = R$. Under these weights, expected next stock price equals today's price grown at the cash rate. Dividing by cash growth makes the stock a martingale. We call the resulting probability law $\mathbb{Q}$. "Risk-neutral" describes this pricing property, not a claim that investors are indifferent to risk or that $\mathbb{Q}$ estimates observed frequencies.

For a general replicated payoff, discounted portfolio gains must have zero conditional mean under such a law. Its discounted current value is therefore the conditional expectation of its discounted payoff. This implication needs the integrability conditions recorded below and in the technical appendix.

6.2 No arbitrage and martingale measures

An arbitrage is an admissible self-financing zero-initial-capital strategy with nonnegative terminal gain almost surely and strictly positive gain with positive probability. In a finite-state, finite-date frictionless market with a strictly positive numeraire and strictly positive physical probabilities, absence of arbitrage is equivalent to existence of an equivalent probability $\mathbb{Q}$ under which discounted traded gains are martingales. Completeness, meaning every contingent payoff is replicable, is equivalent to uniqueness of that measure in this setting.

The finite-state idea is separation. The linear space of zero-cost attainable terminal gains must not intersect the nonnegative cone except at zero. A separating positive state-price vector annihilates attainable zero-cost gains. Normalize its values by the numeraire to obtain equivalent probabilities. Conversely, under such probabilities a zero-cost gain has zero expectation; a nonnegative gain that is positive with positive probability cannot have zero expectation.

With dividends, the relevant discounted gain is

$$\frac{S_t}{B_t} + \int_0^t B_u^{-1} \, \mathrm{d}D_u, \tag{6.1}$$

not merely S_t/B_t. If a European payoff H_T is replicated and integrable under $\mathbb{Q}$, its value is

$$V_t = B_t \mathbb{E}^{\mathbb{Q}}[B_T^{-1} H_T \mid \mathcal{G}_t]. \tag{6.2}$$

In an incomplete market multiple $\mathbb{Q}$'s may be compatible with traded spot and cash. Option quotes restrict the set but need not identify all dynamic risks. A choice of $\mathbb{Q}$ is then a pricing-model selection, not a theorem determining physical expected returns.

6.3 From the pricing expectation to the formula

In the constant-volatility laboratory the pricing stock drift is the cash rate minus dividend yield. Solving the resulting lognormal transition makes the call expectation two truncated normal integrals. We evaluate them rather than present the closed formula as a new axiom.

Under the auxiliary pricing dynamics $\mathrm{d}S = (r - \delta)S\,\mathrm{d}t + \sigma S\,\mathrm{d}W^{Q}$, apply (4.2) to $\log S$:

$$\log S_T = \log s + (r - \delta - \sigma^2/2)\tau + \sigma\sqrt{\tau}Z, \quad Z \sim \mathcal{N}(0,1), \quad \tau = T - t. \tag{6.3}$$

For a European call, $c^E = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[(S_T - K)_+ \mid S_t = s]$. Let

$$d_1 = \frac{\log(s/K) + (r - \delta + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}. \tag{6.4}$$

Derivation: the two truncated expectations

The exercise event is $Z > -d_2$. The cash-strike expectation is $K\mathbb{P}(Z > -d_2) = K\Phi(d_2)$. For the stock term, complete the exponent: $e^{az - a^2/2}\varphi(z) = \varphi(z - a)$, with $a = \sigma\sqrt{\tau}$. Its truncated integral equals $se^{(r-\delta)\tau}\Phi(d_1)$. Therefore

$$c^E = se^{-\delta\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2). \tag{6.5}$$

The European put follows from the pointwise identity $(s - K)_+ - (K - s)_+ = s - K$: $p^E = c^E - se^{-\delta\tau} + Ke^{-r\tau}$. Both formula and parity concern European auxiliary contracts with the specified dividend convention.

What we gained

- Pricing weights interpreted as a martingale probability law.
- The roles of no arbitrage, replication and market completeness.
- The Black–Scholes call formula and European put–call parity.

Where this goes next

We now have two probabilities on future market paths. Their relationship must be explicit before comparing physical forecasts with prices or changing drift coefficients in a volatility model.

Chapter 7

Why P and Q solve different problems

What problem are we solving?

A physical law supplies likelihoods of outcomes; a pricing law supplies weights consistent with traded values. We develop the change of weights first in a finite example, then explain the continuous-time drift change.

7.1 A change of probability as reweighting

On a finite outcome set, suppose $p_i > 0$ are physical probabilities and $q_i > 0$ pricing probabilities. Define $Z_i = q_i/p_i$. Then

$$\sum_i q_i Y_i = \sum_i p_i Z_i Y_i, \qquad \sum_i p_i Z_i = 1.$$

The variable Z gives each physical outcome a new weight. It does not change which outcomes exist. Equivalence means that the two laws have the same zero-probability events. The density process below is this likelihood ratio conditional on information arriving through time.

7.2 A density process

On a fixed finite horizon, let $\mathbb{Q}$ be equivalent to $\mathbb{P}$: they have the same null events. The Radon–Nikodym derivative $Z_T^{\mathbb{Q}} = \mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}$ is strictly positive with $\mathbb{E}^{\mathbb{P}} Z_T^{\mathbb{Q}} = 1$. Its density process is $Z_t^{\mathbb{Q}} = \mathbb{E}^{\mathbb{P}}[Z_T^{\mathbb{Q}} \mid \mathcal{F}_t]$. For integrable Y, conditional Bayes' rule gives

$$\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t] = \frac{\mathbb{E}^{\mathbb{P}}[Z_T^{\mathbb{Q}} Y \mid \mathcal{F}_t]}{Z_t^{\mathbb{Q}}}. \tag{7.1}$$

To verify it, multiply by $Z_t^{\mathbb{Q}} \mathbf{1}_A$, $A \in \mathcal{F}_t$, take $\mathbb{P}$-expectations, and use the defining Radon–Nikodym identity on both sides. It is not valid to replace a $\mathbb{Q}$-expectation by an unweighted $\mathbb{P}$-expectation.

7.3 Why the exponential density has a compensator

For a single normal shock $Z \sim \mathcal{N}(0, h)$, the weight $\exp(-\lambda Z - \lambda^2 h/2)$ has expectation one. Completing the square in its product with the normal density shows that the reweighted shock has mean $-\lambda h$ and variance h. Thus reweighting changes drift while leaving the local noise scale fixed. Girsanov's

theorem extends this calculation to an adapted sequence of infinitesimal weights; a true-martingale condition ensures their total mass remains one.

7.4 Girsanov's drift change

Let W^P be a d-dimensional standard Brownian motion and $\lambda_t \in \mathbb{R}^d$ predictable. Define

$$Z_t^{\mathbb{Q}} = \exp\left\{-\int_0^t \lambda_u^\top \, \mathrm{d}W_u^P - \frac{1}{2}\int_0^t \|\lambda_u\|^2 \, \mathrm{d}u\right\}. \tag{7.2}$$

Assume Novikov's sufficient condition $\mathbb{E}^{\mathbb{P}}[\exp(\frac{1}{2}\int_0^T \|\lambda_u\|^2 \, \mathrm{d}u)] < \infty$. Then $Z^{\mathbb{Q}}$ is a positive true martingale and defines an equivalent measure. Girsanov's theorem states that $W_t^Q = W_t^P + \int_0^t \lambda_u \, \mathrm{d}u$ is Brownian under $\mathbb{Q}$ [14]. Its sign matters: $\mathrm{d}W^P = \mathrm{d}W^Q - \lambda \, \mathrm{d}t$. Substituting into a diffusion $\mathrm{d}X = b^P \, \mathrm{d}t + A \, \mathrm{d}W^P$ gives

$$b^Q = b^P - A\lambda. \tag{7.3}$$

Quadratic covariation is pathwise and remains $AA^\top \, \mathrm{d}t$. Equivalent changes of measure alter drift, not the observed pathwise diffusion covariance.

In the baseline stock model $\mathrm{d}S = \mu S \, \mathrm{d}t + \sigma S \, \mathrm{d}W^P$, discounted cum-dividend gains are martingales when $\lambda = (\mu - r + \delta)/\sigma$. Then the pricing drift is $r - \delta$. A deterministic bounded λ satisfies the stated condition; a fitted unbounded risk-price process needs a separate martingale argument.

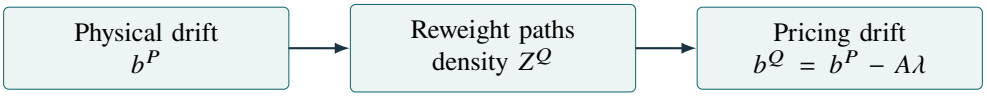


Figure 7.1: The pathwise covariance stays fixed. A change of measure is not permission to choose an unrelated diffusion.

7.5 Risk premia and legitimate comparisons

The pricing kernel combines discounting with the probability weight. Expanding its product expectation into a mean and covariance shows how a nonzero physical expected gain can compensate exposure to states with high pricing weight.

For a European payoff H_T, define the pricing kernel $M_{t,T} = B_t Z_T^{\mathbb{Q}}/(B_T Z_t^{\mathbb{Q}})$. Then $V_t = \mathbb{E}^{\mathbb{P}}[M_{t,T} H_T \mid \mathcal{F}_t]$. For deterministic rates, $\mathbb{E}^{\mathbb{P}}[M_{t,T} \mid \mathcal{F}_t] = D(t,T)$, so

$$V_t = D(t,T)\mathbb{E}^{\mathbb{P}}[H_T \mid \mathcal{F}_t] + \mathrm{Cov}^{\mathbb{P}}(M_{t,T}, H_T \mid \mathcal{F}_t). \tag{7.4}$$

Hence the physical expected payoff minus financed price equals $-\mathrm{Cov}^{\mathbb{P}}(M_{t,T}, H_T \mid \mathcal{F}_t)/D(t,T)$. Expected gains can compensate exposure to states with high marginal pricing weight. A difference between $\mathbb{P}$ and $\mathbb{Q}$ alone is not a mispricing theorem.

Define integrated realized variance $\mathrm{RV}_{t,T} = \int_t^T v_u \, \mathrm{d}u$ for a continuous diffusion. One sign convention for variance-risk premium is

$$\mathrm{VRP}_{t,T} = \mathbb{E}^{\mathbb{Q}}[\mathrm{RV}_{t,T} \mid \mathcal{G}_t] - \mathbb{E}^{\mathbb{P}}[\mathrm{RV}_{t,T} \mid \mathcal{G}_t]. \tag{7.5}$$

The sign convention must be stated; volatility-risk premium using $\sqrt{\mathrm{RV}/(T-t)}$ is a different nonlinear expectation. With jumps, realized quadratic variation includes squared log-price jumps. A model-free variance-replication argument requires a suitable European option continuum and dynamic terms; it cannot be asserted from raw American SPY quotes.

For a live American contract, comparing $\mathbb{E}^{\mathbb{P}}[c^A_{t+h}]$ with c^A_t omits financing, interim exercise and hedging. The main alpha object will therefore be expected *full policy gain*, not an unqualified difference between a physical payoff expectation and a pricing distribution.

What we gained

- A density ratio derived from finite probability weights.
- The sign and covariance restriction of Girsanov's drift change.
- A pricing-kernel explanation of expected gains and risk premia.

Where this goes next

A price can now be represented by a pricing law. Market participants also represent that same price by an implied volatility; we next treat this as an inverse coordinate, not as a physical forecast.

Chapter 8

Implied volatility as a price coordinate

What problem are we solving?

Different strikes and maturities have prices on different scales. Inverting a familiar pricing formula gives a common coordinate, but the inverse must exist and its contract assumptions must match the question.

8.1 Inverting a one-dimensional map

Fix stock price, strike, expiry, rate and dividend convention. In the European laboratory write the call value as $f(\sigma)$. If this function is strictly increasing, a quoted price c in its range determines one number $\sigma_{\text{imp}} = f^{-1}(c)$. This number encodes that price in the chosen formula. It need not equal the future realized standard deviation of returns.

To measure inverse sensitivity, differentiate $f(\sigma_{\text{imp}}) = c$. This gives $f'(\sigma_{\text{imp}})\,\mathrm{d}\sigma_{\text{imp}} = \mathrm{d}c$. A small derivative means that a small price error can produce a large change in the inferred coordinate. The derivative is called vega; we will study the other price derivatives after constructing the surface.

8.2 Implied volatility is an inverse coordinate

For the European map (6.5), differentiation gives

$$\frac{\partial c^E}{\partial \sigma} = s e^{-\delta\tau}\varphi(d_1)\sqrt{\tau} > 0, \qquad s, K, \sigma, \tau > 0. \tag{8.1}$$

The derivative calculation appears in Appendix E. As $\sigma \downarrow 0$, call value tends to $(se^{-\delta\tau} - Ke^{-r\tau})_+$; as $\sigma \to \infty$, it tends to $se^{-\delta\tau}$. Therefore a price strictly between these limits has a unique positive implied volatility. At the boundaries the inverse is limiting or nonunique in an enlarged convention.

For a small price error δc, the inverse function theorem gives $\delta\sigma_{\text{imp}} \approx \delta c/\nu$, with vega $\nu = \partial_\sigma c^E$. Very small vega makes implied volatility unstable. Bid and ask imply an interval of volatility coordinates, not a perfectly measured scalar.

For American SPY prices, define an American implied parameter by solving

$$c^A_{\text{model}}(\sigma; S, K, T, \text{dividends}, r) = c^A_{\text{market}}.$$

The inverse depends on the stopping and dividend model; flat exercise regions can destroy strict monotonicity and identification. Applying the European inverse directly can misattribute early-exercise value to volatility or produce no admissible root. The main policy fits American prices; any European-equivalent surface is model-dependent auxiliary information.

What we gained

- Implied volatility defined by an inverse pricing map.
- Existence, uniqueness and low-vega conditioning understood.
- Separate European and American inversion conventions.

Where this goes next

One implied volatility describes one quote. Across strike and maturity the coordinates form a surface whose shape must respect payoff inequalities.

Chapter 9

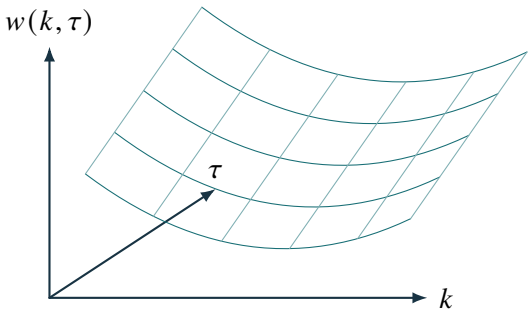
The volatility surface and its restrictions

What problem are we solving?

Fitting every quote separately can create impossible prices between quotes. We organize prices across strikes and maturities, derive the simplest restrictions from payoffs, and then introduce a constrained surface family.

9.1 From a collection of quotes to a surface

A smile is a slice at one expiry; a term structure compares expiries. Changing strike asks which terminal price region a claim pays in. Changing expiry asks how long the claim remains exposed. A surface summarizes these two directions at one observation date. It is not yet a law of motion through observation time. At a fixed observation date, set $\tau = T - t$, forward quote $F_t(T) = S_t e^{\int_t^T (r_u - \delta_u)\,\mathrm{d}u}$ in the continuous-yield auxiliary model, log forward moneyness $k = \log(K/F_t(T))$, and total implied variance $w_t(k,\tau) = \sigma_{\mathrm{imp,t}}(k,\tau)^2\tau$. Smile is variation in strike, skew is a strike derivative, and term structure is variation in maturity. These describe coordinates of prices, not directly the physical law of realized variance.



Strike curvature constrains density; maturity evolution constrains calendar consistency.

Figure 9.1: Schematic total-variance slices. Drawing coordinates only set the geometry; they represent no calibration or empirical surface.

9.2 Static restrictions and what calendar consistency means

A higher strike never increases a European call payoff, so it cannot increase its frictionless price. Convex combinations of strikes give convexity. These elementary inequalities provide the first tests of a proposed surface.

For European calls at fixed maturity and deterministic carry,

$$(S_t e^{-\int \delta} - KD(t,T))_+ \le c^E(K,T) \le S_t e^{-\int \delta}, \quad -D(t,T) \le c^E_K \le 0, \quad c^E_{KK} \ge 0. \tag{9.1}$$

Monotonicity follows from ordering call payoffs by strike; convexity follows from convexity of $(s-K)_+$ in K. A negative butterfly price would purchase a nonnegative payoff for negative cost in the frictionless auxiliary market.

For deterministic proportional dividends, define normalized terminal stock $Y_T = S_T/F_t(T)$ and normalized call $c^E(K,T)/(D(t,T)F_t(T))$ at fixed $k = \log(K/F_t(T))$. The process $S_T e^{-\int_t^T (r-\delta)}$ is a martingale under the selected true-martingale model. Conditional Jensen's inequality implies increasing normalized call value with maturity at fixed normalized strike. In Black–Scholes coordinates, strict positivity of vega makes this equivalent to nondecreasing $w(k,\tau)$.

This is not a blanket claim that raw calls at a fixed numerical strike increase with maturity under every dividend and rate schedule. Discrete cash dividends require direct pricing consistency. For actual American claims with the same payoff definition and nested exercise opportunities, a longer horizon contains the shorter stopping set; the stopping formulation provides the relevant dominance. A fitted European surface is only an auxiliary check on an American pricing model.

9.3 European state-price density

The strike derivative asks how many states still pay when the strike is raised slightly. A second derivative isolates a narrow band of terminal states. Differentiating the payoff expectation makes this interpretation precise.

Assume deterministic rates, an integrable positive S_T, and a continuous $\mathbb{Q}$-density $f^Q_t(s;T)$ conditional on present information. Let $D(t,T) = e^{-\int_t^T r_u \,\mathrm{d}u}$ be the discount factor, distinct from the cumulative dividend process D_t. Then

$$c^E_t(K,T) = D(t,T) \int_K^\infty (s-K) f^Q_t(s;T)\,\mathrm{d}s. \tag{9.2}$$

Differentiating under the integral, justified by integrability and density regularity, gives

$$\partial_K c^E_t = -D(t,T) \int_K^\infty f^Q_t(s;T)\,\mathrm{d}s, \tag{9.3}$$

$$\partial_{KK} c^E_t = D(t,T) f^Q_t(K;T). \tag{9.4}$$

This is the Breeden–Litzenberger relation [4]. With atoms, the second derivative is a measure in the distributional sense. With stochastic rates, it recovers a discounted state-price measure; an ordinary risk-neutral terminal density requires a specified numeraire and forward-measure adjustment.

An American price includes the right to choose τ, so differentiating $c^A(K,T)$ does not isolate a single terminal payoff distribution. If a model writes $c^A = c^E + \mathcal{E}^{\text{early}}$, then $\partial_{KK} c^A = D f^Q + \partial_{KK}\mathcal{E}^{\text{early}}$. The early-exercise term is model-dependent. The main system neither differentiates raw American prices as if European nor calls the resulting curvature a physical forecast.

9.4 A controlled SSVI representation

An unrestricted interpolating curve can violate these restrictions. SSVI chooses an explicit family whose shape and parameter bounds can be checked together. Its parameters describe a surface at a date; they do not by themselves describe future surface moves.

For auxiliary European total variance, a useful structured family is

$$w(k,\theta) = \frac{\theta}{2}\left[1 + \rho_s\phi(\theta)k + \sqrt{(\phi(\theta)k + \rho_s)^2 + 1 - \rho_s^2}\right], \tag{9.5}$$

where $\theta > 0$ is at-the-money total variance, $|\rho_s| < 1$, and $\phi(\theta) > 0$ is smooth. The surface correlation parameter ρ_s is not automatically a spot/variance Brownian correlation.

Read the parameters as shape controls

At $k = 0$, substitution gives $w(0,\theta) = \theta$, so θ sets the level at the forward strike. Differentiation there gives $w_k(0,\theta) = \theta\rho_s\phi(\theta)$, so ρ_s controls the sign of skew and ϕ contributes to its scale. The square-root term bends the wings. These are price-shape controls, not yet coefficients of a stochastic volatility law.

Not every choice yields valid prices. The derivative and wing bounds in Appendix C are sufficient restrictions for this particular family; that appendix also derives the strike-curvature test (C.3). They retain the technical conditions behind the surface without making their verification the reader's first encounter with a smile.

SSVI is a surface coordinate family, not a stochastic law. A time process for its parameters must preserve the restrictions and specify how parameters co-move with spot. The main model may instead keep a nonparametric convex price set or a family of American pricing models; the decision compares their induced future marks.

What we gained

- Strike, maturity and total variance used as price coordinates.
- Static inequalities derived from payoff ordering and convexity.
- State-price density and a constrained SSVI surface family.

Where this goes next

A spread responds to local movements of spot and these surface coordinates. Derivatives of the pricing map quantify those responses; they are the Greeks.

Chapter 10

Greeks as derivatives of a pricing map

What problem are we solving?

A trader needs to know which small market changes affect a position. Multivariable calculus supplies a local exposure map once we specify the price function and which coordinates are held fixed.

10.1 Introduce the sensitivities one at a time

For a price $c(t, s, \sigma)$, first change only the stock coordinate: $c(t, s + \varepsilon, \sigma) - c(t, s, \sigma) = c_s\varepsilon + o(\varepsilon)$. The coefficient c_s is delta, the local number of shares with the same first-order price response. Delta itself changes with s; its derivative c_{ss} is gamma. The second-order contribution to price is $c_{ss}\varepsilon^2/2$.

Next change only volatility. The coefficient c_σ is vega, already encountered when inverting the price. Passage of calendar time at fixed state contributes c_t, called theta. Finally, mixed and second volatility derivatives describe how these local responses interact. Their names merely abbreviate familiar derivatives; the coordinate choices determine their meaning.

10.2 Greeks are coordinate derivatives

For a smooth continuation-region pricing map $c(t, s, \sigma)$, define

$$\Delta = c_s, \quad \Gamma = c_{ss}, \quad \nu = c_\sigma, \quad \Theta = c_t, \quad \text{Vanna} = c_{s\sigma}, \quad \text{Volga} = c_{\sigma\sigma}. \tag{10.1}$$

Theta is a calendar-time derivative at fixed state, so it is minus a remaining-maturity derivative when other coordinates are fixed. Vega is with respect to volatility, not variance; if $v = \sigma^2 > 0$, then $c_v = c_\sigma/(2\sigma)$. The choice of coordinates changes exposure numbers and must be consistent across legs.

For the auxiliary BSM call, differentiation of (6.5) gives

$$\Delta = e^{-\delta\tau}\Phi(d_1), \qquad \Gamma = \frac{e^{-\delta\tau}\varphi(d_1)}{s\sigma\sqrt{\tau}}, \tag{10.2}$$

$$\nu = se^{-\delta\tau}\varphi(d_1)\sqrt{\tau}, \qquad \text{Vanna} = -e^{-\delta\tau}\varphi(d_1)\frac{d_2}{\sigma}, \tag{10.3}$$

$$\text{Volga} = \nu\frac{d_1 d_2}{\sigma}. \tag{10.4}$$

$$\Theta = -\frac{se^{-\delta\tau}\varphi(d_1)\sigma}{2\sqrt{\tau}} + \delta se^{-\delta\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2). \tag{10.5}$$

Appendix E derives the cancellations. American Greeks are derivatives of the American functional, generally obtained from its stopping problem. Near the exercise boundary smooth derivatives may not exist, and small quote or dividend changes can move the boundary.

For a surface state $\zeta = (\zeta_1, \dots, \zeta_d)$, define $\nu_\ell = \partial_{\zeta_\ell} c$, cross derivatives $c_{s\zeta_\ell}$, and Hessian $c_{\zeta_\ell\zeta_k}$. A parallel implied-volatility shift is one direction in this space. Vega-neutral under that shift does not imply neutral to skew or term-structure changes.

10.3 Taylor expansion and Itô P&L

For a finite small move, use Taylor's theorem with a remainder. For a continuous diffusion, retain the quadratic covariations because squared Brownian increments accumulate. These are two related expansions with different error statements, and neither replaces exact repricing across a jump.

For a finite state move $(\Delta t, \Delta s, \Delta\zeta)$, smoothness through third spatial derivatives on the connecting region gives

$$\begin{aligned}\Delta c = c_t \Delta t + c_s \Delta s + \nabla_\zeta c^\top \Delta\zeta + \frac{1}{2} c_{ss} (\Delta s)^2 \\ + \Delta s\, c_{s\zeta}^\top \Delta\zeta + \frac{1}{2} \Delta\zeta^\top c_{\zeta\zeta} \Delta\zeta + \mathcal{R}.\end{aligned} \tag{10.6}$$

The remainder contains third-order spatial terms and mixed time terms; with bounded required derivatives it is $O(\|(\Delta s, \Delta\zeta)\|^3 + |\Delta t|\|(\Delta s, \Delta\zeta)\| + \Delta t^2)$. For diffusion increments, Itô's formula retains quadratic covariations:

$$\begin{aligned}\mathrm{d}c = c_t\, \mathrm{d}t + c_s\, \mathrm{d}S + \nabla_\zeta c^\top\, \mathrm{d}\zeta + \tfrac{1}{2} c_{ss}\, \mathrm{d}[S] \\ + \sum_\ell c_{s\zeta_\ell}\, \mathrm{d}[S, \zeta_\ell] + \tfrac{1}{2} \sum_{\ell,k} c_{\zeta_\ell\zeta_k}\, \mathrm{d}[\zeta_\ell, \zeta_k].\end{aligned} \tag{10.7}$$

Here $[X, Y]$ denotes quadratic covariation. The coefficient of the mixed term has no extra one-half because the symmetric Hessian contributes two equal cross terms.

Delta is first-order stock exposure, gamma is curvature exposure to squared stock moves, vega is first-order surface motion, theta is passage of time, vanna is stock/surface co-movement and volga is surface curvature. Their monetary contributions require actual state changes; a label alone is not a risk measure. Large moves, exercise, dividends, jumps and surface-coordinate changes require complete revaluation or explicit jump terms.

What we gained

- Delta, gamma, vega and theta developed from separate coordinate changes.
- Mixed sensitivities and their dependence on the chosen surface coordinates.
- A finite Taylor expansion distinguished from continuous Itô gains.

Where this goes next

A portfolio adds the sensitivities of its legs. We next solve small hedge equations before using a general exposure matrix or pseudoinverse.

Chapter 11

Portfolio exposure geometry

What problem are we solving?

Naming a spread does not specify which risks it removes. We first solve for holdings that cancel a chosen exposure, then identify the degrees of freedom available for other objectives.

11.1 Two options and a stock hedge

Let two option contracts have deltas d_1, d_2 and vegas v_1, v_2, with the contract multipliers already included. Choose option quantities q_1, q_2 and h stock shares. To remove vega and delta, solve

$$v_1 q_1 + v_2 q_2 = 0, \qquad d_1 q_1 + d_2 q_2 + h = 0.$$

If $v_2 \neq 0$, set $q_1 = a$, giving

$$q_2 = -\frac{v_1}{v_2} a, \qquad h = -\left(d_1 - d_2 \frac{v_1}{v_2}\right) a.$$

The free scalar a scales a direction that preserves both hedge equations. Gamma and nonparallel surface exposure can remain. The desired gain, costs and capital must decide whether to use this direction and at what size.

11.2 From a line of solutions to a nullspace

Collect any chosen exposure equations as $Gq = b$. Each row is one risk coordinate and each column one contract. If q_0 reaches the target, every other solution differs from q_0 by a vector annihilated by G. That is the nullspace. A pseudoinverse finds a particular solution only after we know that the target lies in the range of the exposure map.

11.3 The exposure map and its nullspace

Let $G \in \mathbb{R}^{p \times m}$ have columns equal to contract sensitivities in p chosen risk coordinates, including multipliers m_j. For option vector q, exposures are Gq. If stock shares h are included, augment with a column $g_S = (1, 0, \ldots, 0)^\top$ for pure spot delta in these coordinates. Then $\widetilde{G}(q^\top, h)^\top = Gq + g_S h$.

To impose $Gq = b$, let the compact singular value decomposition be $G = U_r D_r V_r^\top$, where $r = \operatorname{rank} G$, positive singular values form D_r, and U_r, V_r have orthonormal columns. A solution

exists exactly when $b \in \operatorname{range} G$, equivalently $U_r U_r^\top b = b$. The minimum Euclidean-norm solution is

$$q_0 = G^+ b = V_r D_r^{-1} U_r^\top b. \tag{11.1}$$

If $N_G \in \mathbb{R}^{m\times(m-r)}$ has orthonormal columns spanning $\ker G$, every solution is $q = q_0 + N_G z$, $z \in \mathbb{R}^{m-r}$. Indeed $G(q-q_0) = 0$; orthogonality of $q_0 \in \operatorname{range} G^\top$ and $N_G z$ gives $\|q\|^2 = \|q_0\|^2 + \|z\|^2$, proving minimality.

For the weighted errors below, $H_G \succ 0$ means $z^\top H_G z > 0$ for every nonzero z, with H_G symmetric. The weaker notation $H_G \succeq 0$ permits zero in some directions. Thus a positive definite weight gives an actual squared error length.

Technical note: an unstable or incompatible hedge

Small singular values amplify target error by division in D_r^{-1}. An overdetermined incompatible system has no exact hedge. A weighted least-squares fit minimizes $(Gq-b)^\top H_G (Gq-b)$, $H_G \succ 0$, possibly with an inventory penalty and bounds. It leaves an explicit residual exposure. Removing a small singular direction changes what risk is controlled and must be reported as a tolerance.

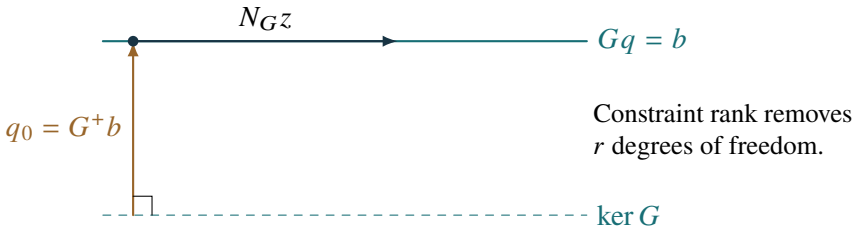


Figure 11.1: An affine Greek constraint set. The schematic shows one remaining degree of freedom; actual option space may be high-dimensional.

11.4 Neutrality is local and model-dependent

If $G_t q = 0$, then after a state move $G_{t+h} q = (G_{t+h} - G_t) q$, generally nonzero. A zero delta does not remove gamma. A nullspace based on one model need not be a nullspace for another. For a model family G^M, robust neutrality is $\sup_M \|G^M q - b\| \le \epsilon_G$. In the limit of exact neutrality for all models, the feasible intersection may collapse to zero.

The analogy to futures factors is precise but limited: both annihilate rows of a loading/exposure map. Futures factor gains are linear in the stated return coordinates; option Greeks linearize a nonlinear pricing functional at one state. Option payoff and assignment risk persist outside that tangent approximation.

What we gained

- A symbolic hedge solved directly without a pseudoinverse.
- Range, rank and nullspace interpreted as exposure feasibility and freedom.
- A local hedge distinguished from protection against full nonlinear losses.

Where this goes next

The Greeks describe responses at a state, but the constant-volatility laboratory gives too rigid a law for future states. We next make volatility itself random.

Chapter 12

Why volatility becomes a state

What problem are we solving?

A single fixed volatility cannot describe changing uncertainty or independent moves in option prices at unchanged stock price. We add a variance state and ask what must be known now to predict its future.

12.1 A scalar random variance before a diffusion model

Imagine tomorrow's variance can be either v_L or v_H, both positive. Even if tomorrow's stock price equals today's, the option's future value can differ between these variance states. Stock price alone is therefore not a sufficient state. The pair (S, v) may be sufficient if its transition law depends only on the current pair.

This is the Markov-state idea: knowing the present state makes the earlier history unnecessary for the model's future conditional law. It is a property of the specified dynamics, not a claim that real markets forget their history. A model may recover the property by enlarging its state.

12.2 From random variance to stochastic volatility

With a positive variance process v_t, write

$$\mathrm{d}S_t/S_t = \mu_t\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{S,P}.$$

The square root converts variance per unit time to a standard-deviation scale. This stock equation does not define a full model: we still need a law for v_t and its co-movement with the stock shock. For example, correlated stock and variance shocks can make a stock decline coincide with a rise in uncertainty. An option hedge against the stock shock need not cancel a second, independent variance shock.

12.3 Why an unconstrained Gaussian variance is awkward

A Gaussian noise model for v can assign positive probability to negative variance. Exponentiating a Gaussian state is one solution; using a diffusion coefficient that shrinks near zero is another. Heston uses the latter route: its variance diffusion has coefficient proportional to $\sqrt{v}$. We will derive its moments and state its boundary behavior before discussing its pricing advantage.

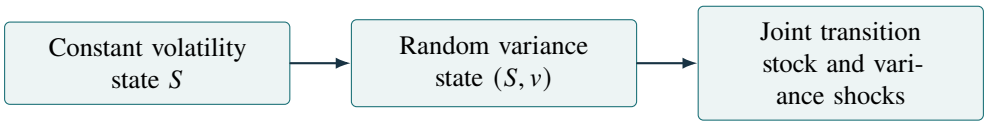


Figure 12.1: The new state is introduced because future option values can differ at the same stock price.

What we gained

- A reason variance must sometimes be retained as a separate state.
- The Markov property interpreted as a sufficiency requirement.
- The missing variance and correlation specifications identified.

Where this goes next

Heston provides a concrete positive-variance transition. It also shows precisely why pricing calibration and physical forecasting need different drift information.

Chapter 13

Heston: a first dynamic volatility model

What problem are we solving?

We need a variance model that remains nonnegative, has a tractable transition structure and permits stock–variance co-movement. Heston meets these modeling aims with a two-dimensional Markov state.

13.1 Read the variance equation before solving it

The drift $\kappa(\theta - v)$ pulls variance toward $\theta > 0$ at speed $\kappa > 0$, much as an OU drift pulls a scalar state toward its mean. The noise coefficient $\xi\sqrt{v}$ introduces a new positive constant ξ, the volatility of variance. Its dependence on v changes boundary behavior: unlike an additive Gaussian shock, the diffusion vanishes at zero. The precise positivity statement is given below.

The physical and pricing laws use the same local diffusion coefficients if they are equivalent measures on one process. Their variance drifts can differ. We therefore keep a measure subscript on mean-reversion parameters rather than quietly substituting calibrated values for physical estimates.

13.2 Heston dynamics and boundary conditions

To obtain the conditional mean, take expectations in the variance equation and solve a linear differential equation. For variance, apply Itô to the square: the extra squared-diffusion term is the source of the additional coefficient in the second-moment equation.

Under either stated measure $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$, the variance process is

$$\mathrm{d}v_t = \kappa_M(\theta_M - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{v,M}, \qquad \kappa_M, \theta_M, \xi > 0. \tag{13.1}$$

Spot has drift μ_t under $\mathbb{P}$ and $r_t - \delta_t$ under $\mathbb{Q}$, volatility $\sqrt{v_t}$, and Brownian correlation $\mathrm{d}\langle W^{S,M}, W^{v,M}\rangle_t = \rho\,\mathrm{d}t$. The square-root process has a nonnegative solution. If $v_0 > 0$ and $2\kappa_M\theta_M \geq \xi^2$, the Feller condition makes zero inaccessible; if it fails, zero can be reached, although the process remains nonnegative with its specified boundary behavior. This is not a condition guaranteeing every moment of an option price.

Taking expectations and solving the linear differential equation gives

$$m(h) = \mathbb{E}^M[v_{t+h} \mid v_t = v] = \theta_M + (v - \theta_M)e^{-\kappa_M h}. \tag{13.2}$$

Derivation: variance from the second moment

For the second moment, Itô gives $\mathrm{d}(v^2) = [(2\kappa_M\theta_M + \xi^2)v - 2\kappa_M v^2]\,\mathrm{d}t + 2\xi v^{3/2}\,\mathrm{d}W^v$. Localize to justify expectation and use finite moments. The integrating-factor solution, minus $m(h)^2$, yields

$$\mathrm{Var}^M(v_{t+h} \mid v_t = v) = \frac{v\xi^2}{\kappa_M}e^{-\kappa_M h}(1 - e^{-\kappa_M h}) + \frac{\theta_M\xi^2}{2\kappa_M}(1 - e^{-\kappa_M h})^2. \tag{13.3}$$

The stationary density is Gamma with shape $2\kappa_M\theta_M/\xi^2$ and rate $2\kappa_M/\xi^2$. Appendix D derives it from zero probability flux. Its mean and variance are θ_M and $\theta_M\xi^2/(2\kappa_M)$, consistent with the limits above.

13.3 Two drivers and unspanned volatility

A stock and cash can hedge one stock shock. Write the variance shock as a correlated stock component plus an independent component to see which part remains unspanned. The stock martingale condition fixes only the first market price of risk.

Write independent Brownian drivers $W^{1,P}, W^{2,P}$, and

$$\mathrm{d}S_t/S_t = \mu_t\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{1,P}, \tag{13.4}$$

$$\mathrm{d}v_t = b_v^P(v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\left(\rho\,\mathrm{d}W_t^{1,P} + \sqrt{1-\rho^2}\,\mathrm{d}W_t^{2,P}\right), \tag{13.5}$$

with $|\rho| < 1$. The stock martingale restriction fixes $\lambda_1 = (\mu - r + \delta)/\sqrt{v}$ where $v > 0$. It does not determine λ_2, because cash and stock do not span the independent variance driver. Under $\mathbb{Q}$,

$$b_v^Q(v) = b_v^P(v) - \xi\sqrt{v}\{\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2\}. \tag{13.6}$$

Define $\lambda_v = \rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2$. If $b_v^P = \kappa_P(\theta_P - v)$ and $\lambda_v = (a_v + b_v v)/(\xi\sqrt{v})$, then

$$\kappa_Q = \kappa_P + b_v, \qquad \kappa_Q\theta_Q = \kappa_P\theta_P - a_v. \tag{13.7}$$

Here a_v, b_v are risk-price coefficients, not the drift map $b_v^P(\cdot)$. Positivity of κ_Q, θ_Q, boundary behavior and the density-martingale condition must be checked. The singularity at $v = 0$ makes this an assumption-sensitive parameterization.

Fitting unrelated Heston diffusions under $\mathbb{P}$ and $\mathbb{Q}$ with different ξ or ρ does not establish an equivalent measure change of one process. It instead defines two approximations. Such pairs may be used as model uncertainty, but the conceptual difference must be retained.

13.4 A generator packages the Itô drift

For a smooth test function of $(\log S, v)$, Itô's formula produces first derivatives from drift and second derivatives from shock covariance. The operator collecting these deterministic drift terms is called the generator. Applying it to an exponential test function is convenient because derivatives return multiples of the same function. The affine variance coefficients then close the calculation in two scalar functions of time.

13.5 Correlation, characteristic transforms and pricing

The mixed quadratic covariation is $\mathrm{d}\langle S, v\rangle = \rho\xi v S\,\mathrm{d}t$. If $\rho < 0$, downward spot shocks tend to coincide with higher variance shocks, generating asymmetric future spot distributions in the model.

This mechanism can produce skew, but the actual surface shape depends on all parameters and horizon.

For deterministic r, δ, let $X = \log S$. Its $\mathbb{Q}$-generator acting on smooth $f(x, v)$ is

$$\mathcal{L}^{Q} f = (r - \delta - v/2) f_x + \kappa_Q(\theta_Q - v) f_v + \tfrac{1}{2} v f_{xx} + \rho\xi v f_{xv} + \tfrac{1}{2}\xi^2 v f_{vv}. \tag{13.8}$$

The conditional characteristic transform $\mathbb{E}^{\mathbb{Q}}[e^{iuX_T} \mid X_t = x, v_t = v]$, $u \in \mathbb{R}$, admits the ansatz $e^{iux + A(\tau,u) + B(\tau,u)v}$.

Derivation: why the exponential ansatz closes

Here $i = \sqrt{-1}$ and u is a real Fourier frequency. The characteristic function is the expected complex wave e^{iuX_T}; under the usual inversion conditions these waves encode the distribution. For the proposed exponential, $f_x = iuf$, $f_{xx} = -u^2 f$, $f_v = Bf$, $f_{vv} = B^2 f$ and $f_{xv} = iuBf$. Divide the generator equation by f and collect the constant and v coefficients. This is why two ordinary differential equations suffice. In the displayed autonomous equations take r, δ constant on the interval; deterministic time-varying carry enters at the corresponding calendar time in the backward equation. Substitution into $\partial_\tau f = \mathcal{L}^{Q} f$ and division by f give

$$\partial_\tau B = \tfrac{1}{2}\xi^2 B^2 + (\rho\xi iu - \kappa_Q) B - \tfrac{1}{2}(u^2 + iu), \tag{13.9}$$

$$\partial_\tau A = iu(r - \delta) + \kappa_Q \theta_Q B, \qquad A(0, u) = B(0, u) = 0. \tag{13.10}$$

This Riccati structure is Heston's tractability advantage [10]. Transform inversion prices suitable European payoffs subject to integrability and branch consistency. American valuation still requires stopping with state (S, v); a European transform is not an American closed form.

What we gained

- A nonnegative mean-reverting variance model and its conditional moments.
- A decomposition of spanned and unspanned volatility shocks.
- The generator and Riccati transform structure behind Heston pricing.

Where this goes next

The model equations are known, but current variance and their coefficients usually are not. We next separate inference from historical observations and calibration to current option quotes.

Chapter 14

Physical forecasting and pricing calibration

What problem are we solving?

Historical observations tell us about future market transitions; today's quotes constrain pricing values at the current state. These are different data-to-model maps, even when they share a variance state.

14.1 Observe a noisy proxy before introducing a belief

Suppose $v_t > 0$ is latent and a positive variance proxy Y_t satisfies $\log Y_t = \log v_t + \epsilon_t$, with centered Gaussian error of variance $r > 0$. This is a declared observation model, not an identity for squared returns. If $p_t^-(v)$ is a density predicted from earlier observations, then Bayes' rule gives

$$p_t(v \mid Y_t) = \frac{\ell(Y_t \mid v) p_t^-(v)}{\int_0^\infty \ell(Y_t \mid u) p_t^-(u)\, \mathrm{d}u}.$$

The likelihood ℓ says how probable this reading is at a candidate variance. The denominator normalizes. Only after this scalar step do we need the term *belief*: it is the conditional distribution over the latent state.

14.2 Add unknown parameters, then alternative models

If ϑ contains unknown physical drift and variance coefficients, carry a joint density $p_t(v, \vartheta)$. Each candidate coefficient gives a different transition and likelihood; integrate over the posterior rather than treat the best fit as certain. A fixed parameter is held fixed along a future path. A later model label will add another uncertainty layer only after we have constructed alternative dynamics.

14.3 The calibration problem at one date

Fix one candidate pricing model and a current state. Calibration asks which pricing parameters produce values inside today's bid–ask intervals. Those parameters are restricted by the $\mathbb{P}$-to-$\mathbb{Q}$ relations if the two laws describe one process. A price fit cannot, by itself, supply a history-based estimate of physical drift. The American pricing functional used below means the value of the right to exercise before expiry; its stopping construction is developed in Chapter 18. For European claims the same inverse-problem discussion uses the terminal-payoff expectation already derived.

14.4 Calibration is a separate inverse problem

Here z_t denotes the current pricing state, initially (S_t, v_t); $\mathcal{V}_j^{A,M}$ is the cash value of contract j in pricing model M; and ϑ^Q is its pricing-parameter vector. The superscript A records the American exercise right, not a different probability law.

At date t, define the quote-consistent pricing set

$$\Theta_t^{Q,M} = \{\vartheta^Q : \ c_{j,t}^{\text{bid}} \le \mathcal{V}_j^{A,M}(t, z_t; \vartheta^Q)/m_j \le c_{j,t}^{\text{ask}} \text{ for every eligible } j\}. \tag{14.1}$$

Allow a declared error tolerance if exact feasibility is not maintained. If the set is empty, reject that exact-fit specification or widen only a justified observation-error model. A least-squares calibration may minimize weighted quote discrepancies plus a regularization penalty over valid parameters, but its optimizer is an estimate, not uniquely identified truth.

Let J_Q be the Jacobian of contract marks with respect to pricing parameters. Small singular values of J_Q mean substantially different parameters produce almost the same prices. Their future hedging implications can still differ. The physical likelihood, quote likelihood and a prior/ambiguity set together decide how much confidence to place in those directions. Current quote fitting provides no direct estimate of the physical market prices of risk without historical restrictions.

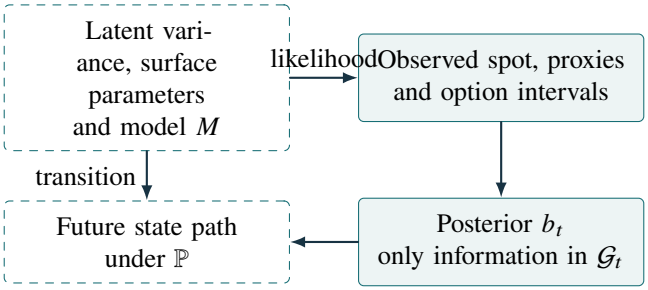


Figure 14.1: Observation and latent state. The posterior integrates quote noise, physical parameter uncertainty and pricing-model ambiguity.

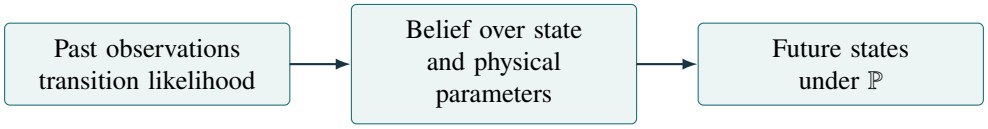


Figure 14.2: Historical inference supplies a transition forecast. Quote calibration constrains a pricing map at a date.

What we gained

- A scalar noisy-variance update before general filtering notation.
- Separate uncertainty about the state and the fixed parameters.
- A quote-consistent pricing set and a local identification diagnostic.

Where this goes next

Heston restricts how a whole surface can fit. Local volatility takes a different route: infer a state-dependent diffusion directly from a smooth European price surface.

Chapter 15

Local volatility and the surface inverse problem

What problem are we solving?

Can a one-dimensional diffusion reproduce an entire family of European option prices? Local volatility answers this question by making the stock's noise coefficient a deterministic function of time and stock price.

15.1 What local means

In $\sigma_{\text{loc}}(t, S_t)$, the function is fixed after calibration, but its value along a random stock path is random. At the same date and stock price it always gives the same instantaneous variance. Unlike Heston, it has no additional variance state that can move independently of stock.

15.2 Why a forward density equation appears

European call prices average terminal payoffs against a terminal stock density. To connect maturity derivatives of call prices to a local diffusion, we need to know how that density changes with maturity. The forward equation below expresses conservation of probability: drift transports density, while diffusion spreads it. Integration against the call payoff transfers its derivatives to the payoff and turns the equation into a relation between observable surface derivatives and the unknown diffusion coefficient.

15.3 Local volatility and Dupire's relation

Consider the European auxiliary $\mathbb{Q}$-diffusion $\mathrm{d}S_t = (r_t - \delta_t)S_t\,\mathrm{d}t + \sigma_{\text{loc}}(t, S_t)S_t\,\mathrm{d}W_t^{\mathcal{Q}}$, with deterministic rates/yields and a smooth positive density $p(T, s)$. The forward density equation is

$$\partial_T p = -\partial_s[(r_T - \delta_T)sp] + \tfrac{1}{2}\partial_{ss}[\sigma_{\text{loc}}^2(T, s)s^2 p]. \tag{15.1}$$

Derivation: integrate the density equation against a call

Differentiate $c(K, T) = D(0, T)\int_K^\infty (s-K)p(T, s)\,\mathrm{d}s$ with respect to T, substitute the density equation, and integrate by parts with vanishing boundary terms. The drift integral is $(r_T - \delta_T)D\int_K^\infty sp\,\mathrm{d}s =$

$(r_T - \delta_T)(c - Kc_K)$; the diffusion integral is $\frac{1}{2}K^2\sigma_{\text{loc}}^2 c_{KK}$. Together with discount differentiation,

$$\sigma_{\text{loc}}^2(T,K) = \frac{2[c_T + (r_T - \delta_T)Kc_K + \delta_T c]}{K^2 c_{KK}}. \tag{15.2}$$

The denominator must be strictly positive, and the numerator nonnegative. Price-surface derivatives must exist and boundary integrals must be controlled. Noisy quotes and nearly zero density make the inverse ill-conditioned [6].

For a regular stochastic-volatility model, conditioning its forward density equation on spot suggests $\sigma_{\text{loc}}^2(T,s) = \mathbb{E}^{\mathbb{Q}}[v_T \mid S_T = s]$, under the assumptions permitting the marginal projection. Thus a local-volatility model can match European marginal distributions without matching joint spot/variance paths. It has no independent variance driver at a given (t,S). This restriction matters to future skew moves, forward-start exposure and hedge errors.

Calibrating local volatility to today's continuum does not estimate physical drift or tomorrow's calibration process. For SPY, one must first obtain a defensible European-equivalent surface or calibrate the American model directly; (15.2) applied to American prices is not valid.

What the inverse has identified

In (15.2), the denominator is strike curvature multiplied by K^2; it measures how much terminal probability occupies that strike region. The numerator combines maturity change with deterministic carry corrections. Dividing by a nearly empty density region amplifies quote errors. Even a well-conditioned inverse identifies this particular diffusion representation, not a unique physical mechanism for future surface changes.

What we gained

- Local volatility distinguished from an independently random variance state.
- Dupire's formula derived through the forward density equation.
- The European scope and conditioning limitations of surface inversion.

Where this goes next

Local volatility and stochastic volatility can agree on present European values while disagreeing on future joint paths. We now isolate why a snapshot does not determine a movie.

Chapter 16

Why today's surface does not determine tomorrow's

What problem are we solving?

A surface at one date constrains terminal distributions at different maturities. Trading before expiry and hedging across dates require how those future states are joined, which is additional information.

16.1 The missing joint law: a finite example

Let X_1, X_2 each be equally likely to be -1 or 1. In one model take $X_2 = X_1$; in another, take X_1 and X_2 independent. Both models have the same marginal law at each date, so every expectation of a function of one date agrees. But

$$\mathbb{E}[X_2 \mid X_1] = X_1 \quad \text{in the first model,} \qquad \mathbb{E}[X_2 \mid X_1] = 0 \quad \text{in the second.}$$

This example illustrates marginal versus joint information; it is not asserted to be a pair of no-arbitrage stock models. A financial model must additionally respect martingale, positivity and contract constraints. The logical gap remains: single-date marginals alone do not specify conditional transitions.

16.2 What a present option surface determines

In the regular European setting, strike curvature recovers each maturity's pricing marginal. It does not determine the joint spot–variance path, the physical probability of surface shifts, or the co-movement needed for hedge losses. The local-volatility projection discussed previously is a concrete way to match marginals while imposing different dynamics from a model with an independent variance driver.

If a rule sells a spread tomorrow, its gain depends on tomorrow's surface at tomorrow's spot, not simply on today's value of a later-expiry claim. A daily recalibration sequence is itself an observed time series. Describing it requires a transition model and an observation-error convention.

16.3 What state must be retained?

For fixed Heston parameters, current (S_t, v_t) determines the model's future law. Some volatility models require more of the past. This is a reason to enlarge the state, not a reason to abandon

conditioning. We next develop a weighted history model and ask precisely which past information determines future variance.

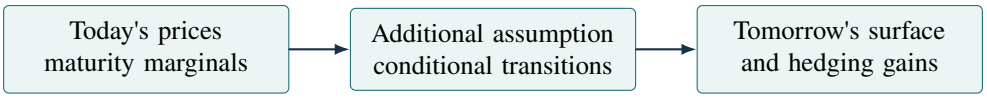


Figure 16.1: The middle arrow cannot be recovered from a surface snapshot alone.

What we gained

- A symbolic distinction between marginal and joint laws.
- A precise limit on what present European prices identify.
- The state-sufficiency question that motivates volatility memory.

Where this goes next

Rough volatility is introduced next as one particular history-dependent model. Its purpose, extra state requirements and comparison with Heston can now be stated before its kernel is written.

Chapter 17

Rough volatility and the information in a history

What problem are we solving?

A single variance value may not summarize how recent shocks affect different future horizons. A weighted history can distinguish paths that end at the same variance but imply different future curves.

17.1 Why an exponential kernel has a short state description

Consider $Z_t = \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s$. For $u > t$, split the integral at t:

$$Z_u = e^{-\kappa(u-t)} Z_t + \int_t^u e^{-\kappa(u-s)}\,\mathrm{d}W_s.$$

Every past shock's future weight is its current weight times the same factor $e^{-\kappa(u-t)}$. Thus one number Z_t summarizes the past. This is the mechanism behind an OU state and its exponential memory.

17.2 A kernel that cannot factor this way

For a power kernel $k(t-s)$, the ratio $k(u-s)/k(t-s)$ generally depends on s. Two histories with the same present weighted sum can therefore give different future weighted sums. We need the collection of past contributions at future horizons, not just today's variance. "Memory" here means this need to retain weighted history. It is distinct from positive long-lag correlation.

The exponent $H \in (0, 1/2)$ controls short-time regularity in the model below. A smaller H makes short increments decay more slowly as the time step shrinks. Appendix D develops fractional Brownian motion and proves the regularity comparisons; the model's actual driver is the Volterra integral introduced next, which is not the same Gaussian process.

17.3 A Volterra Gaussian driver

For $0 < H < 1/2$, define

$$\mathcal{K}_H(t-s) = \sqrt{2H}(t-s)^{H-1/2}, \quad s < t, \qquad Y_t = \int_0^t \mathcal{K}_H(t-s)\,\mathrm{d}W_s^{v,Q}. \tag{17.1}$$

The kernel is square integrable because $\int_0^t 2H(t-s)^{2H-1}\,\mathrm{d}s = t^{2H} < \infty$. Itô isometry gives

$$\mathrm{Cov}^{\mathbb{Q}}(Y_t, Y_u) = 2H \int_0^{t\wedge u} (t-s)^{H-1/2}(u-s)^{H-1/2}\,\mathrm{d}s, \qquad \mathrm{Var}^{\mathbb{Q}}(Y_t) = t^{2H}. \tag{17.2}$$

This Riemann–Liouville-type driver has local roughness related to H, but is not the stationary-increment fractional Brownian motion in (D.2). Confusing the two would change the conditional covariance and forecasting law.

17.4 A representative rough Bergomi specification

Exponentiating the Gaussian driver ensures positive variance. The deterministic correction in the exponent normalizes its expectation to the chosen forward-variance curve. This is the same completing-the-square identity used for lognormal stock values.

Let $\xi_0(t) > 0$ be a deterministic initial forward-variance curve, locally bounded and integrable, and let $\eta > 0$ be volatility-of-volatility. Under a selected $\mathbb{Q}$, define

$$v_t = \xi_0(t) \exp\left\{\eta Y_t - \frac{\eta^2}{2} t^{2H}\right\}, \tag{17.3}$$

$$\mathrm{d}S_t/S_t = (r_t - \delta_t)\,\mathrm{d}t + \sqrt{v_t}\left(\rho\,\mathrm{d}W_t^{v,\mathcal{Q}} + \sqrt{1-\rho^2}\,\mathrm{d}W_t^{\perp,\mathcal{Q}}\right), \tag{17.4}$$

where $W^{v,\mathcal{Q}}$ and $W^{\perp,\mathcal{Q}}$ are independent Brownian motions and $|\rho| < 1$. A centered Gaussian Z with variance a satisfies $\mathbb{E}e^{\eta Z} = e^{\eta^2 a/2}$, by completing the normal exponent. Hence

$$\mathbb{E}^{\mathbb{Q}}[v_t] = \xi_0(t), \qquad \mathbb{E}^{\mathbb{Q}}[v_t^p] = \xi_0(t)^p \exp\left\{\frac{p(p-1)\eta^2}{2} t^{2H}\right\} \tag{17.5}$$

for each fixed real p with positive $\xi_0(t)$. Finite variance moments do not by themselves prove the discounted stock stochastic exponential is a true martingale. That property is an additional admissibility check on the selected pricing model; otherwise expectation pricing can lose the intended forward.

Correlation gives $\mathbb{E}[v_t v_u] = \xi_0(t)\xi_0(u)\exp\{\eta^2 \mathrm{Cov}(Y_t, Y_u)\}$. Therefore

$$\mathrm{Cov}(v_t, v_u) = \xi_0(t)\xi_0(u)\left(e^{\eta^2 \mathrm{Cov}(Y_t, Y_u)} - 1\right). \tag{17.6}$$

This covariance is tied to a Volterra kernel rather than the single exponential decay of stationary Heston variance. Neither is universally more accurate; the empirical question concerns the horizons and observables used by the policy.

17.5 Conditional forward variance

To condition on today, divide every future driver into the integral over the observed past and the integral over future shocks. The first is known under full information; the second is Gaussian with a computable variance. Integrating only the second part produces the forward-variance curve.

For $u > t$, split

$$Y_u = \underbrace{\int_0^t \mathcal{K}_H(u-s)\,\mathrm{d}W_s^{v,\mathcal{Q}}}_{A_t(u)} + \underbrace{\int_t^u \mathcal{K}_H(u-s)\,\mathrm{d}W_s^{v,\mathcal{Q}}}_{B_{t,u}}. \tag{17.7}$$

Conditional on $\mathcal{F}_t$, $A_t(u)$ is known and $B_{t,u}$ is centered Gaussian with variance $\int_t^u \mathcal{K}_H(u-s)^2\,\mathrm{d}s$. The conditional Gaussian exponential contributes one half of its future variance in the exponent. Subtracting the original full-horizon compensator leaves exactly the compensator for the past integral. Gaussian integration yields

$$\xi_t(u) := \mathbb{E}^{\mathbb{Q}}[v_u \mid \mathcal{F}_t] = \xi_0(u)\exp\left\{\eta A_t(u) - \frac{\eta^2}{2}\int_0^t \mathcal{K}_H(u-s)^2\,\mathrm{d}s\right\}. \tag{17.8}$$

For fixed u, ordinary Itô calculus in $t < u$ gives $\mathrm{d}\xi_t(u) = \eta\mathcal{K}_H(u-t)\xi_t(u)\,\mathrm{d}W_t^{v,Q}$. The negative compensator cancels the Itô correction. This is a martingale in the conditioning time t.

Knowing only v_t determines one scalar weighted history Y_t, not the collection $A_t(u)$ for every $u > t$. Two past paths can give the same current variance and different forward-variance curves. The state needed for exact prediction is therefore a history or a curve. An American pricing problem under (17.3) is naturally a stopping problem on such a state, not merely on (S_t, v_t).

For two future dates $u, v > t$, conditional covariance of Y_u, Y_v is $\int_t^{u\wedge v} \mathcal{K}_H(u-s)\mathcal{K}_H(v-s)\,\mathrm{d}s$. Together with the conditional means $A_t(\cdot)$, this determines the Gaussian future driver law. It produces the joint variance path law required by option repricing and hedging, rather than only a single expected variance.

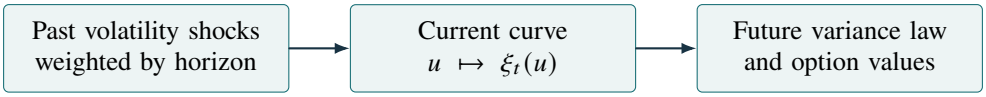


Figure 17.1: The curve retains horizon-dependent information that a single current variance cannot retain.

17.6 Physical forecasting is an additional specification

Equation (17.8) is a $\mathbb{Q}$-conditional expectation. To forecast actual future variance under $\mathbb{P}$, specify a physical model and its risk-price relation. If $\mathrm{d}W_s^{v,Q} = \mathrm{d}W_s^{v,P} + \lambda_s^v\,\mathrm{d}s$, then

$$Y_t^Q = \int_0^t \mathcal{K}_H(t-s)\,\mathrm{d}W_s^{v,P} + \int_0^t \mathcal{K}_H(t-s)\lambda_s^v\,\mathrm{d}s. \tag{17.9}$$

The second term is a history-dependent drift contribution. Setting it to zero is a zero-volatility-risk-price assumption, not a consequence of roughness. An equivalent measure change preserves local path regularity; it does not freely change the Hurst exponent of the same path law.

Observed option prices can restrict $\xi_t^Q(\cdot)$, while historical volatility proxies inform physical dynamics. Both are noisy and only partially observed. Apparent roughness of a measurement sequence can also arise from observation error. A valid test separates latent roughness from the noise model and compares forecasts at declared sampling horizons.

17.7 Finite-dimensional approximation and its error

A sum of exponentials approximates the power kernel and converts its history into finitely many OU-type coordinates. The isometry measures the driver error directly; transferring that error to prices or exercise decisions is a further analytical problem.

The kernel has a Laplace representation. With $\alpha = 1/2 - H \in (0, 1/2)$ and gamma function $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1}e^{-y}\,\mathrm{d}y$, substitution $y = xt$ gives

$$t^{-\alpha} = \frac{1}{\Gamma(\alpha)}\int_0^\infty e^{-xt}x^{\alpha-1}\,\mathrm{d}x. \tag{17.10}$$

Approximating this integral by positive weighted exponentials gives $\mathcal{K}_H(t) \approx \sum_{\ell=1}^{L} \omega_\ell e^{-\kappa_\ell t}$. Define states $Z_{\ell,t} = \int_0^t e^{-\kappa_\ell (t-s)}\, \mathrm{d}W_s^v$. They satisfy $\mathrm{d}Z_{\ell,t} = -\kappa_\ell Z_{\ell,t}\, \mathrm{d}t + \mathrm{d}W_t^v$, and $Y_t \approx \sum_\ell \omega_\ell Z_{\ell,t}$. All lifted factors share a Brownian driver and are correlated.

The driver approximation error has exact mean-square value

$$\mathbb{E}|Y_t - Y_t^{(L)}|^2 = \int_0^t \left| \mathcal{K}_H(u) - \sum_{\ell=1}^{L} \omega_\ell e^{-\kappa_\ell u} \right|^2 \mathrm{d}u. \tag{17.11}$$

Thus selecting a lift is a kernel-approximation problem with an explicit error metric. Controlling this error does not automatically bound an American exercise boundary or tail hedging loss; those require stability and integrability arguments for the pricing/control functional. Any finite lift is Markovian in a larger state, while the limiting kernel generally requires infinitely many modes.

17.8 Why short-maturity skew enters the comparison

A power kernel permits a stronger short-maturity skew response than a regular finite-dimensional volatility diffusion. This is one motivation for testing rough models, not evidence that a particular observed series must be rough. Appendix D retains the perturbative scaling calculation and its restrictions. For trading, the comparison must ultimately concern future marks and financed hedge losses at the policy's actual horizons.

17.9 Model comparison tied to decisions

Property	BSM	Heston	Local volatility	Rough Bergomi
State for pricing	S	(S, v)	(t, S)	S and history/curve
Variance risk	deterministic	independent driver	spot-determined	Volterra driver
Gaussian driver	Brownian	Brownian	Brownian	Gaussian Volterra plus Brownian
Memory	none beyond state	exponential variance covariance	through spot	kernel-weighted history
European tractability	closed formula	affine transform	forward/backward PDE	path expectation
American value	stopping	stopping	stopping	stopping on enlarged state
Surface dynamics	rigid	parameter-restricted	determined by local map	curve-dependent
Short-time skew	zero at fixed σ	bounded in regular case	depends on local regularity	singular order can occur

Table 17.1: Qualitative restrictions, not empirical rankings. All entries refer to the maintained versions developed in the text.

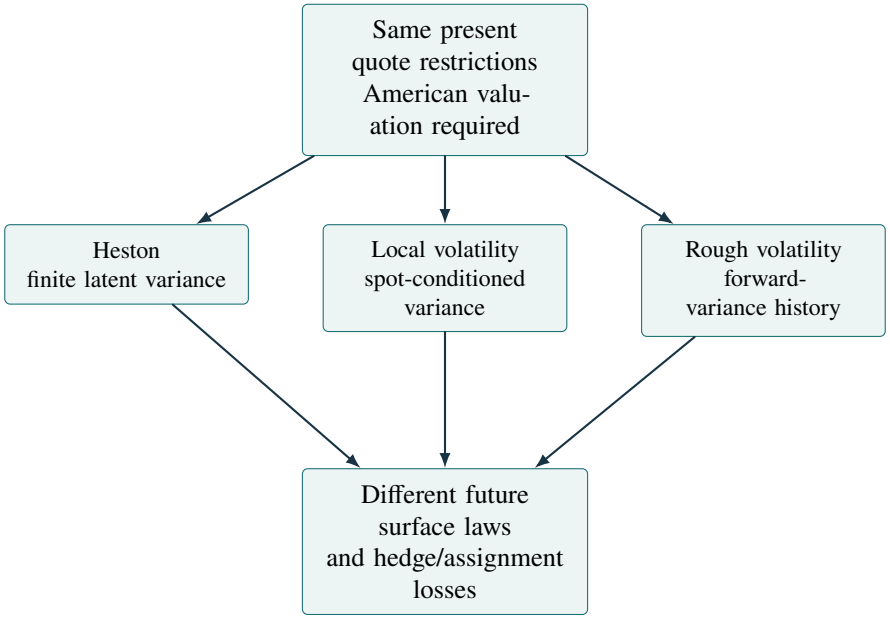


Figure 17.2: Model comparison concerns the future gain distribution needed by the trade. Present fit alone cannot select the physical dynamics.

What we gained

- A direct explanation of why power-kernel history needs an enlarged state.
- The rough Bergomi variance law and conditional forward-variance curve.
- Finite-dimensional lifts, error bounds and a qualified model comparison.

Where this goes next

We now have pricing dynamics on a simple or enlarged state. American exercise uses those dynamics to choose a stopping time; its recursion starts with a two-date exercise decision.

Chapter 18

American exercise as optimal stopping

What problem are we solving?

An American holder may take the exercise value now or retain a valuable future choice. A terminal-payoff expectation ignores that choice, so we must optimize over allowed exercise times.

18.1 Two exercise dates

Suppose exercise is allowed today and tomorrow. The immediate payoff is $g(S_t)$; waiting gives tomorrow's payoff discounted under the chosen pricing law. The value is therefore

$$c_t^A = \max\left\{g(S_t), \mathbb{E}^{\mathbb{Q}}\left[\frac{B_t}{B_{t+1}} g(S_{t+1}) \mid \mathcal{G}_t\right]\right\}.$$

The maximum compares cash values at the same date. On three dates, tomorrow's payoff in this expression is replaced by tomorrow's optimally chosen value. Repeating that substitution yields the backward recursion below. A stopping time formalizes the rule that exercise must be decided using information available when it occurs.

18.2 Stopping times and the Snell envelope

A random time τ is a stopping time for $(\mathcal{F}_u)$ when $\{\tau \le u\} \in \mathcal{F}_u$ for each u; exercise cannot depend on information arriving after the decision. Let $\mathcal{T}_{t,T}$ contain allowed stopping times between t and T. For a selected pricing model $\mathbb{Q}$, the American per-share value is

$$c_t^A = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[\frac{B_t}{B_\tau} g(S_\tau) \mid \mathcal{F}_t\right]. \tag{18.1}$$

The essential supremum is the least almost-sure upper bound over conditional random values. If the market state is Markov, this can be written $c^A(t, z)$; under partial information it is a value on the belief state. The continuous-time attainment and integrability conditions are recorded in Appendix A; the finite-grid recursion below exhibits the basic decision directly.

On a discrete exercise grid, backward conditioning gives

$$c_N^A = g(S_N), \qquad c_j^A = \max\left\{g(S_j), \mathbb{E}^{\mathbb{Q}}\left[\frac{B_j}{B_{j+1}} c_{j+1}^A \mid \mathcal{F}_j\right]\right\}. \tag{18.2}$$

This is the smallest discounted supermartingale dominating discounted exercise value, the Snell envelope. The maximum follows because the holder can exercise now or retain the right to choose optimally later. It also proves $c^A \geq c^E$.

18.3 Continuation and exercise regions

At a state where waiting is optimal, the pricing PDE applies. At a state where exercise is optimal, value equals the exercise payoff. The obstacle inequality combines these alternatives; its signs encode that the holder can always take the payoff and cannot improve value by a suboptimal delay.

Let $\mathcal{L}^Q$ be the spatial generator of the pricing diffusion, defined by its Itô drift acting on smooth functions. In a continuation region, waiting preserves value, so $c_t + \mathcal{L}^Q c - rc = 0$. In the exercise region $c = g$. Combining them gives the variational inequality

$$\max\{g - c,\ c_t + \mathcal{L}^Q c - rc\} = 0, \qquad c(T, z) = g(z). \tag{18.3}$$

Thus $c \geq g$ and the continuation operator is nonpositive globally where derivatives exist. The free boundary separates the two regions and is part of the solution, not prescribed beforehand. Smooth pasting $c_s = g_s$ holds under regular diffusion and boundary assumptions; jumps, degeneracy or nonsmooth rewards can invalidate a naive use of it. In those cases the stopping formulation remains primary.

There is no general elementary closed form for the American value with stochastic volatility and discrete dividends. The mathematical task is an optimal stopping problem or an associated variational inequality with terminal and dividend boundary conditions.

18.4 Discrete dividends and early exercise

At a deterministic ex-dividend time t_d with known cash dividend $d_d > 0$, the idealized price jump is $S_{t_d} = S_{t_d-} - d_d$, assuming the resulting price remains positive. The discounted gain process includes the dividend, so the share's cum-dividend value does not jump under this idealization. Just before the event,

$$c^A(t_d-, s) = \max\{g(s), c^A(t_d+, s - d_d)\}. \tag{18.4}$$

This shows directly why retaining a call can sacrifice access to an imminent dividend. For a put, immediate receipt of strike cash and financing also matter. Under nonnegative rates and no dividends, the usual frictionless American call argument can rule out early exercise; that argument does not apply unchanged to SPY's dividends.

Unknown future dividends require a dividend state and conditional law, not a deterministic adjustment guessed after the event. Short-position assignment is controlled by other holders and allocation conventions. It is modeled as a conditional event with inventory consequences (20.9), not as our own optimal stopping decision. Optimal stopping values and actual assignment intensities are related but not identical objects.

A diffusion has support arbitrarily near zero, so subtracting a fixed cash dividend globally can create an inadmissible negative share value. All fixed-dividend formulas above are restricted to $s > d_d$. To close the mathematical model on the remaining paths, specify a nonnegative payout rule, for example $d_d(s) = \min(d_d, s)$, with the stock absorbed at zero if all equity is distributed. Trading then enters its absorbing liquidation state; outstanding claims retain their stated payoff obligations. This is a model boundary convention, not an assertion about an observed SPY event. Its probability and losses belong to model stress, and a different corporate/distribution boundary model is a challenger.

Mathematical checkpoint

Can a spread have bounded expiry payoff and still create an unaffordable intermediate stock position? Write the assignment transition of only one short leg before inspecting the terminal payoff.

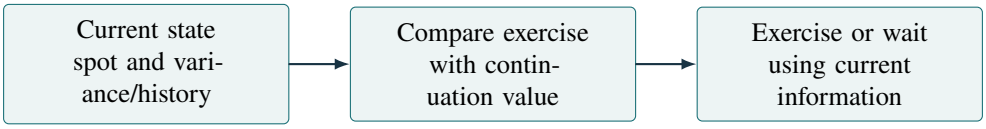


Figure 18.1: For rough volatility, continuation depends on the enlarged state. For a short position, the holder's decision becomes an external assignment event.

What we gained

- A two-date exercise decision extended to the Snell envelope.
- Continuation and exercise regions with a derived obstacle inequality.
- Dividend jumps, state requirements and the distinction between exercise and assignment.

Where this goes next

A trading system can now reprice every leg at future states and include early exercise. We combine that gain law with desired exposures and solve for a spread.

Chapter 19

Constructing spreads from forecasts and exposures

What problem are we solving?

The exposure equations describe what a spread controls; the physical gain law describes why one might hold it. We now connect these maps, incorporate model uncertainty, and choose actual contract quantities.

19.1 Assemble the information in a deliberate order

Start with current stock price and a belief over variance. Add parameter uncertainty as in Chapter 14. For each alternative model, keep the state it actually needs: a Heston variance, a local-volatility stock state, or a rough forward curve. Only now add model uncertainty. A model label is a choice of transition and pricing mechanism, not another observed market quote.

The general notation below collects these objects after their separate development. Its purpose is to produce the law of the gain from a proposed portfolio and holding rule, including the possibility of early exercise.

19.2 A joint state and observation model

Let $z_t = (\log S_t, v_t, \zeta_t, \mathfrak{d}_t, \ell_t)$ contain spot, variance, admissible surface or risk-price coordinates ζ_t, dividend information $\mathfrak{d}_t$ and liquidity state ℓ_t. For rough models replace v_t by the necessary forward-variance/history state or a declared finite lift. These state components inhabit a product of Euclidean spaces and, when needed, a function space.

A model M specifies a physical transition kernel $K^{P,M}_{\vartheta^P}(z, \mathrm{d}z')$ and a pricing transition/stopping model under $\mathbb{Q}^M$. If they are claimed to describe equivalent measures on the same process, impose the measure-change conditions from Chapter 7, including dynamics for risk-price state ζ. If daily surface coordinates are merely revalued as frozen parameters, call this a quasi-static repricing approximation and include its dynamic inconsistency in model error. Static no-arbitrage at every date alone does not establish absence of dynamic arbitrage.

Let observed vector Y_t include spot increments, noisy variance proxies and option prices. A possible measurement equation is

$$Y_t^{\mathrm{opt}} = \mathcal{V}^{A,M}(t, z_t; \vartheta^{\mathcal{Q}}) + \epsilon_t^{\mathrm{quote}}, \tag{19.1}$$

where $\mathcal{V}^{A,M} \in \mathbb{R}^m$ consists of contract values and the quote-noise law has specified covariance or bid–ask support. A variance proxy has a separate likelihood; latent v_t is not equal to squared observed return. For example, aggregated squared log increments target integrated variance under a high-frequency continuous-semimartingale limit, while discrete sampling, noise and jumps alter that approximation.

The full belief recursion

Define the belief $b_t(\mathrm{d}z, \mathrm{d}\vartheta, \mathrm{d}M) = \mathbb{P}(z_t, \vartheta, M \in \mathrm{d}(z, \vartheta, M) \mid \mathcal{G}_t)$. With fixed unknown parameter ϑ, prediction and updating are

$$b^-_{t+1}(\mathrm{d}z', \mathrm{d}\vartheta, \mathrm{d}M) = \int K^{P,M}_{\vartheta}(z, \mathrm{d}z') b_t(\mathrm{d}z, \mathrm{d}\vartheta, \mathrm{d}M), \tag{19.2}$$

$$b_{t+1}(\mathrm{d}z', \mathrm{d}\vartheta, \mathrm{d}M) \propto \ell_{\vartheta,M}(Y_{t+1} \mid z') b^-_{t+1}(\mathrm{d}z', \mathrm{d}\vartheta, \mathrm{d}M). \tag{19.3}$$

The proportionality constant is the reciprocal of the positive finite predictive likelihood integral. This is a mathematical estimator of states and parameters, not an instruction to set physical parameters equal to a pricing calibration.

19.3 The predictive law and the gain map

First forecast a future path under the physical law. At each future state, evaluate the selected pricing/stopping functional. Apply the hedge and cash ledger to those marks. The outer probability is physical even though a pricing expectation is used inside each future mark.

For a horizon $h > 0$, define

$$\mathsf{P}^M_t = \mathcal{L}_{\mathbb{P}}(S_{[t,t+h]}, z_{[t,t+h]}, \text{fills, dividends, assignments} \mid \mathcal{G}_t, M). \tag{19.4}$$

It is obtained by integrating the physical transition law against b_t, including parameter uncertainty. Future contract marks are random variables $C^M_{j,u} = \mathcal{V}^{A,M}_j(u, z_u)$. At each realized future state this is a $\mathbb{Q}^M$ pricing/stopping functional; the outer distribution of those states is under $\mathbb{P}$.

These model marks are not guaranteed executable liquidation prices. When the observation specification includes quote discrepancy, its future conditional law must also be integrated: the predicted market mark is $C^{\mathrm{obs}}_{j,u} = C^M_{j,u} + \epsilon^{\mathrm{quote}}_{j,u}$, with eligible bid–ask bounds and the stated noise convention. Actual liquidation uses market prices in the ledger. Treating model marks as exact is the explicit zero-discrepancy specialization, not an unrecorded source of gain.

For options held without exercise until $t + h$, and a prescribed stock hedge h_u, define the financed gain

$$\begin{aligned}\Pi^{q,\pi,M}_{t,t+h} = {} & q^\top C^M_{t+h} - \frac{B_{t+h}}{B_t} q^\top C^{\mathrm{entry}}_t \\ & + B_{t+h} \int_t^{t+h} B_u^{-1} h_u (\mathrm{d}S_u + \mathrm{d}D_u - r_u S_u \mathrm{d}u) - \mathcal{K}^{\pi}_{t,t+h}.\end{aligned} \tag{19.5}$$

Here $\mathcal{K}$ is the terminal-valued cost and extra funding ledger, and C^{entry}_t represents actual cash entry prices with any cost component counted there excluded from $\mathcal{K}$. Hedge purchase and sale principal is financed through self-financing stock gains. If options exercise or trades alter q, use the full ledger of Chapter 22; (19.5) is the no-exercise fixed-inventory specialization.

The alpha vector in a linear one-period approximation is the vector of physical expected financed per-contract gains. The exact decision quantity is

$$\alpha_t(q, \pi; M) = \mathbb{E}^{\mathbb{P}}[\Pi^{q,\pi,M}_{t,t+h} \mid \mathcal{G}_t]. \tag{19.6}$$

If each leg uses a different future valuation convention while sharing a common spot path, the portfolio may be internally inconsistent. Each model must revalue all legs jointly.

19.4 What the forecast is claiming

The main relative-value forecast concerns joint changes in spot, variance level, skew and maturity structure, expressed by z. A choice to predict only realized variance is a restricted submodel; one cannot then claim an independent forecast of vol-of-vol or future skew. Surface motions can dominate a spread's marked gain before expiry even when realized variance is forecast correctly.

For a variance-like payoff, VRP from Chapter 7 is one possible state variable. Its usefulness depends on whether the actual spread and hedge approximate that payoff after American effects, dividends, costs and financing. An empirical edge claim compares forecast-conditioned gains against a matched risk-compensation benchmark, not merely the sign of a pricing-versus-physical variance difference.

Let $m_M = \mathbb{E}^{\mathbb{P}}[\Pi \mid \mathcal{G}_t, M]$ and $s_M^2 = \mathrm{Var}^{\mathbb{P}}(\Pi \mid \mathcal{G}_t, M)$. Posterior model averaging with weights ω_M gives $\mathrm{Var}(\Pi \mid \mathcal{G}_t) = \sum_M \omega_M s_M^2 + \sum_M \omega_M (m_M - \bar{m})^2$, $\bar{m} = \sum_M \omega_M m_M$. Within each model an analogous decomposition separates parameter and process uncertainty. A robust policy instead takes the worst allowed model; it does not require assigning arbitrary posterior probabilities to incompatible specifications.

Mathematical checkpoint

When a predicted gain is positive under Heston and negative under a rough-volatility challenger, is the disagreement in physical forecasts, future pricing marks, hedge behavior, or assignment? Which observation can distinguish those explanations?

19.5 Choose a criterion before adding every constraint

For one candidate spread direction $q = az$, a simple mean–variance criterion is expected financed gain minus a multiple of its variance. The scalar a chooses size; exposure equations determine z. Tail, funding and ambiguity terms are added only to express separate preferences or restrictions. In the full criterion below, cash costs already belong to the gain and must not be subtracted a second time as penalties.

19.6 The objective and feasible decisions

Let $q \in \mathbb{Z}^m$ be the post-trade option vector and h the stock hedge. Let π denote a prescribed subsequent hedge/exercise policy, or include it in the joint optimization. For model M and statistical parameter law u in ambiguity set $\mathcal{U}_M$, let $L^{M,u} = -\Pi_{t,t+H}^{q,h,\pi,M,u}$ be the full financed loss over declared horizon H.

One complete robust objective is

$$\max_{(q,h,\pi)\in\mathcal{A}_t} \inf_{M\in\mathcal{M},\ u\in\mathcal{U}_M} \Big\{ \mathbb{E}^{\mathbb{P}}[\Pi^{q,h,\pi,M,u} \mid \mathcal{G}_t] - \lambda_{\mathrm{tail}}\,\mathrm{ES}_\alpha(L^{M,u} \mid \mathcal{G}_t) - \tfrac{\gamma}{2}\,\mathrm{Var}^{\mathbb{P}}(\Pi^{q,h,\pi,M,u} \mid \mathcal{G}_t) - \lambda_{\mathrm{con}}\|D_{\mathrm{con}} q\|_2^2 - \lambda_{\mathrm{asg}}\mathcal{R}_{\mathrm{asg}}^{M,u}(q,h) \Big\}. \quad (19.7)$$

All penalty weights are nonnegative. Expected shortfall is $\mathrm{ES}_\alpha(L) = \inf_z\{z + (1-\alpha)^{-1}\mathbb{E}[(L-z)_+]\}$; its definition remains valid with assignment atoms [13]. Costs, liquidity slippage and financing are already in Π. The last terms are explicit preference penalties, not additional cash expenses.

For example, $\mathcal{R}_{\text{asg}}$ can be expected positive shortfall of immediate funding after assignment, given a defined reserve: it quantifies a cash constraint failure distinct from mark variance. The diagonal matrix D_{con} penalizes concentration across contracts or grouped maturity exposures. One may omit variance if tail utility better represents preferences; (19.7) states a criterion rather than claiming all risk penalties are simultaneously necessary.

The feasible set includes

$$\begin{aligned} |h + G_\Delta q| &\le \epsilon_\Delta, & |G_\nu q - \nu^\star| &\le \epsilon_\nu, & |G_\Gamma q| &\le \bar{\Gamma}, \\ |q_j| &\le \bar{q}_j, & |h| &\le \bar{h}, & \mathfrak{m}_t(q, h, z_t) &\le M_t^{\text{avail}}, \\ X_t^{\text{post}} &\ge X_t^{\text{reserve}}, & \sup_{\omega \in \mathcal{S}_{\text{stress}}} L(q, h; \omega) &\le L_{\max}. \end{aligned} \tag{19.8}$$

Here $G_\Delta, G_\nu, G_\Gamma$ are selected rows of G; $\nu^\star$ expresses the desired volatility direction; ϵ's are declared hedge tolerances; $\mathfrak{m}_t$ is the applicable conservative margin map; and the stress set includes spot/surface shocks and single-leg assignments. Maturity eligibility imposes $q_j = 0$ for excluded expiries and bounds signed or gross exposure by expiry bucket. Cash after trade is given by (1.2). Reserve and margin feasibility must also be checked after stress transitions.

The model set $\mathcal{M} = \{M_{\text{BSM}}, M_{\text{Heston}}, M_{\text{local}}, M_{\text{rough}}, \ldots\}$ contains only declared admissible variants with the relevant American correction/stopping model. A disagreement measure is $\mathfrak{D}_t(q) = \sup_M \alpha_t(q; M) - \inf_M \alpha_t(q; M)$. If zero is feasible and all nonzero choices have a nonpositive worst-case criterion relative to zero, the optimizer does not initiate a spread. A model-fragile apparent gain is thus rejected by the decision itself.

19.7 Existence, convex approximations and integer limits

With bounded integer q, there are finitely many option vectors. For a fixed admissible hedge-policy family with compact parameter set, bounded h, compact ambiguity sets and continuous expected loss/risk functionals with uniform integrability, the upper-semicontinuous robust objective attains a maximum on a nonempty compact feasible set. General dynamic policies require the Bellman existence conditions discussed later. If zero does not satisfy a mandated nonzero vega target, feasibility must be checked; changing that target requires the declared fallback rule.

In an affine one-period gain approximation $\Pi = q^\top R - c(q - q^-)$, expected gain is linear, covariance risk is quadratic and expected shortfall of affine loss is convex. Convex costs and a convex continuous feasible set give a concave maximization problem; taking the infimum of concave model criteria preserves concavity. Model-dependent nonlinear assignment, state-dependent hedging and integer counts generally remove this simplification. Their solutions require the actual discrete/stochastic optimization, not a closed-form claim.

Continuous Greek equations can be solved in nullspace coordinates $q = q_0 + N_G z$. In an unconstrained mean-variance residual problem, maximizing $a^\top q - \frac{\gamma}{2} q^\top \Sigma_R q$ gives

$$z^\star = (\gamma N_G^\top \Sigma_R N_G)^{-1} N_G^\top (a - \gamma \Sigma_R q_0), \tag{19.9}$$

provided the reduced covariance is positive definite. If it is singular, apply a declared robust penalty or explicit bounds. Actual contracts require an integer feasible solution and tolerated Greek residuals; rounding alone is not a proof of feasibility.

19.8 Payoff geometry produces names

Two tail conditions on three call holdings leave one degree of freedom. Solving that small linear system produces a butterfly direction; its familiar name follows the algebra. It does not establish a

forecast or financing plan.

First restrict support to two calls with common expiry and $K_1 < K_2$, and require the asymptotic stock slope to vanish. The condition is $q_1 + q_2 = 0$. The remaining ray $q = a(1, -1)$ has payoff $a[(S_T - K_1)_+ - (S_T - K_2)_+]$: a vertical structure emerges. Its sign and amplitude are determined by the criterion and capital constraints, not by its name.

For three calls $K_1 < K_2 < K_3$ at one expiry, impose zero upper-tail slope and zero upper-tail intercept: $\sum_j q_j = 0$ and $\sum_j q_j K_j = 0$. Let $a = K_2 - K_1 > 0$, $b = K_3 - K_2 > 0$. Solving gives the one-dimensional direction

$$q = \zeta \left(1, -\frac{a+b}{b}, \frac{a}{b}\right)^\top . \tag{19.10}$$

Below K_1 its payoff is zero; between K_1, K_2 the slope is ζ; between K_2, K_3 it is $-\zeta a/b$; above K_3 it returns to zero. A butterfly is therefore a nullspace of two tail constraints. For equal spacing this specializes to weights proportional to $(1, -2, 1)$, a symbolic identity, not a trading example.

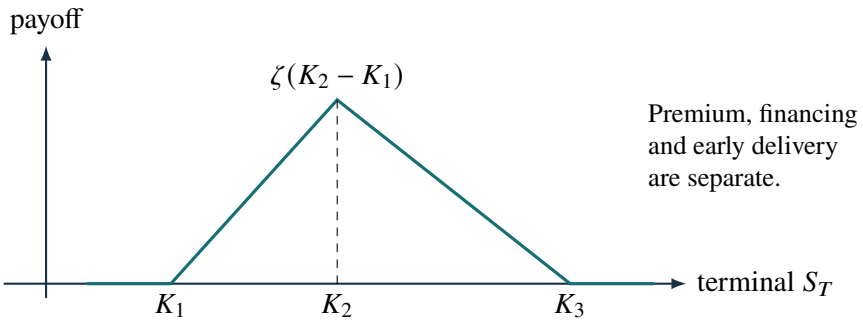


Figure 19.1: Symbolic terminal payoff of a three-strike nullspace direction, shown per unit multiplier. The schematic is not the complete pathwise P&L of American legs.

If support contains the same strike at two maturities, a prescribed vega cancellation solves $\nu_1 q_1 + \nu_2 q_2 = 0$, yielding $q_2 = -(\nu_1/\nu_2)q_1$ when $\nu_2 \neq 0$. This can create a ratio calendar, with stock used to control delta. Different strikes and maturities give a diagonal support. The ratio is determined by exposure geometry and may not be integral. If desired risk is a maturity difference rather than zero total vega, the equality right-hand side changes accordingly.

Names describe support and coefficients after the optimization. A multi-maturity spread has no single terminal payoff diagram until one specifies what happens when the earlier leg expires. Physical assignment can create stock before later legs terminate. The dynamic model, not a static payoff label, defines the complete trade.

What we gained

- A joint physical path forecast composed with future pricing marks.
- A financed gain objective with explicit risk, ambiguity and feasibility terms.
- Spreads derived from exposure/tail equations and then checked for integer feasibility.

Where this goes next

The chosen spread is still a target. Execution, hedge revisions, dividends and assignment determine the holdings that actually earn its gain.

Chapter 20

Execution, hedging, assignment and financing

What problem are we solving?

A hedge may be delayed, legs may fill at different times, and a short contract may turn into stock and strike cash. We write these transitions explicitly before using them in dynamic control.

20.1 Start from the cash account

The trade map in Chapter 1 deducts premiums and stock purchases from cash. Between trades, stock earns price changes and dividends, option marks change, and cash accrues interest. A self-financing hedge purchases its shares with this same account; hedge principal is not an unexplained extra gain. Between trades, exercise and dividend jumps, if X earns rate r and marks are semimartingales, self-financing means

$$\mathrm{d}W_t = r_t X_t \,\mathrm{d}t + h_t(\,\mathrm{d}S_t + \,\mathrm{d}D_t) + q_t^\top \,\mathrm{d}C_t - \,\mathrm{d}K_t^{\mathrm{tr}} - \,\mathrm{d}H_t^{\mathrm{fund}}. \tag{20.1}$$

The last term is any additional funding/borrow charge beyond the baseline rate. If cash has asymmetric rates, replace rX by $r^\ell X^+ - r^b X^-$, with $X^+ = \max(X, 0)$, $X^- = \max(-X, 0)$, and remove overlapping charges from H^{fund}.

20.2 Multi-leg trading frictions

For contract trade a_j, let $s_j = (C_j^{\mathrm{ask}} - C_j^{\mathrm{bid}})/2$ and use midpoint marks only as a reference. A cash cost model is

$$K^{\mathrm{tr}}(a, b) = \sum_j s_j|a_j| + s_S|b| + \tfrac{1}{2}a^\top \Lambda_o a + \eta_S b^2 + K^{\mathrm{fees}}(a, b), \tag{20.2}$$

where $\Lambda_o \succeq 0$, $\eta_S \geq 0$, and s_S is share half-spread. The quadratic terms approximate size-dependent impact. A separate slippage random variable describes execution-price deviation from this model. Actual paid/received cash prices supersede the approximation in the ledger.

Let τ_j be the random fill time of leg j and $A_j(u)$ its cumulative fill process. The intermediate option inventory is $q_u = q_t^- + A(u)$, so $G_u q_u + g_S h_u$ can be large before all legs fill. A legging-risk penalty might be

$$\mathbb{E}^{\mathbb{P}} \int_t^{t+T_e} \|\widetilde{G}_u(q_u, h_u) - b^{\mathrm{target}}\|_{H_G}^2 \,\mathrm{d}u, \qquad \|x\|_{H_G}^2 = x^\top H_G x. \tag{20.3}$$

This is an exposure-duration criterion, not cash cost. A contingent multi-leg instruction can reduce asynchronous fills but has its own fill probability and price distribution. The order control must specify both; targets cannot be treated as guaranteed executions.

20.3 The delta-hedged variance identity

To understand one source of hedged gain, keep the pricing volatility fixed while allowing actual stock quadratic variation to differ. Subtract stock gains and financing from the option change; the stock drift cancels and the remaining gamma term compares realized and priced variance.

In the European smooth baseline, take a fixed option $c(t, S)$ priced with volatility $\widehat{\sigma}$, but suppose actual continuous quadratic variation is $\mathrm{d}[S]_t = v_t S_t^2 \,\mathrm{d}t$. Hold $-c_s$ shares against one option. Its gain in excess of financing is

$$\begin{aligned} \mathrm{d}\Pi &= \mathrm{d}c - c_s(\,\mathrm{d}S + \delta S\,\mathrm{d}t) - r(c - c_s S)\,\mathrm{d}t && (20.4)\\ &= \left[c_t + (r-\delta)Sc_s - rc + \tfrac{1}{2}v_t S_t^2 c_{ss}\right]\,\mathrm{d}t && (20.5)\\ &= \tfrac{1}{2}\Gamma_t S_t^2 (v_t - \widehat{\sigma}^2)\,\mathrm{d}t, && (20.6) \end{aligned}$$

where the final equality uses (4.5). A gamma-weighted realized-versus-priced variance difference appears. Physical spot drift cancels in this continuously delta-hedged idealization.

When the pricing surface moves, the vega, vanna, volga and risk-price terms from (10.6) also enter. American stopping adds exercise transitions. The displayed identity is therefore a diagnostic laboratory, not the complete SPY gain equation.

20.4 Discrete hedge error and transaction costs

Continuous adjustment is an idealization. Holding yesterday's delta over a finite interval leaves a stochastic residual, while adjusting more frequently raises turnover cost. Squared Gaussian increments quantify the first effect locally.

On hedge dates t_k, hold $h_u = -\Delta_{t_k}$ for $u \in (t_k, t_{k+1}]$. Relative to continuous delta hedging, the stochastic gain difference is

$$\mathcal{E}_{\mathrm{disc}} = \sum_k \int_{t_k}^{t_{k+1}} (\Delta_u - \Delta_{t_k})\,\mathrm{d}S_u, \tag{20.7}$$

plus the associated financing/dividend drift differences under the same convention. For a locally constant gamma and diffusion variance, $\Delta_u - \Delta_{t_k} \approx \Gamma_{t_k}(S_u - S_{t_k})$. The centered leading martingale contribution over a step is

$$\frac{\Gamma_{t_k}}{2}\left[(S_{t_{k+1}} - S_{t_k})^2 - v_{t_k} S_{t_k}^2 \Delta t_k\right]. \tag{20.8}$$

For a Gaussian frozen-coefficient increment with variance $a_k = v_{t_k} S_{t_k}^2 \Delta t_k$, the variance of its square is $2a_k^2$. Thus step hedge-error variance is approximately $\Gamma_{t_k}^2 a_k^2/2$. Over a fixed horizon and a regular fine grid, the aggregate diffusion hedge-error variance is of order the step length under bounded coefficients.

Expected absolute delta turnover per step is approximately $|\Gamma| S\sqrt{v}\,\sqrt{2\Delta t/\pi}$. With proportional share cost, total cost over a fixed horizon grows like $(\Delta t)^{-1/2}$, while the preceding variance shrinks like Δt. The balance determines a finite hedge frequency under this local model. Jumps and exercise events do not vanish by this diffusion scaling, and stochastic surface exposures can dominate it.

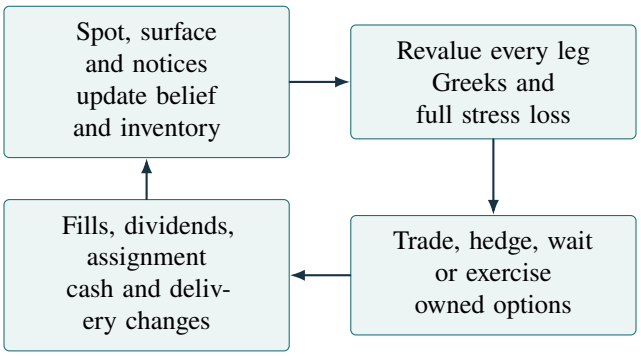


Figure 20.1: The hedge loop includes assignment notices and physical delivery. A static Greek-neutral spread does not remain neutral automatically.

20.5 Exercise is a state transition

For one long call, exercise removes the option, adds its delivered shares and pays strike cash. For a short call, all three signs reverse. We encode both with a signed removal count, then check equity before and after the event.

Let e_j be the signed number of contracts removed by exercise or assignment: $e_j > 0$ for long options exercised and $e_j < 0$ for short contracts assigned. Immediately after exercise is recognized,

$$q'_j = q_j - e_j, \qquad h' = h + \sum_j m_j \chi_j e_j, \qquad X' = X - \sum_j m_j \chi_j K_j e_j. \tag{20.9}$$

For a short call $e_j < 0$, stock inventory falls and cash rises. For a short put, stock inventory rises and cash falls. The long-leg counterpart has opposite signs. A long option elsewhere in the spread is not automatically exercised.

Holding spot and pre-exercise marks fixed, the equity change is

$$W' - W = \sum_j e_j \{m_j \chi_j (S - K_j) - C_j\}. \tag{20.10}$$

For rational in-the-money exercise on a frictionless exercise boundary, $C_j = m_j g_j(S) = m_j \chi_j (S - K_j)$, so the instantaneous change is zero. Exercise away from that boundary can transfer remaining time value; fees and liquidation costs add further loss. Equation (20.10) accounts for the disappearance of the option and the acquired delivery obligation exactly.

Physical settlement has a lag and assignment notices can arrive later than the exercise decision. Let pending stock and cash obligations be $(h^{\text{pend}}, X^{\text{pend}})$. Recognizing exercise places the increments in (20.9) into these pending accounts. Equity includes their marked value; at delivery they transfer into settled accounts without a second gain. Their funding and dividend entitlements depend on the contract and settlement calendar. The policy includes these pending obligations and any uncertainty before notice in its state. The idealized instantaneous map is a compression of this more complete ledger.

An admissible policy observes its information dates, keeps integrable gains and costs, respects collateral/position constraints, and has discounted wealth bounded below by a specified admissibility bound up to liquidation. Integrability is imposed under the measures in which an expectation is taken; it is not inherited automatically from a fitted model.

What we gained

- Actual fill and hedge processes distinguished from target exposures.
- The restricted delta-hedged variance identity and discrete hedge error.
- Signed delivery maps, pending obligations and a consistent financing convention.

Where this goes next

These events alter future choices. Dynamic control must choose orders and owned-option exercise while treating short assignment as a random event outside the trader's control.

Chapter 21

Dynamic control of the option portfolio

What problem are we solving?

Choosing a spread once does not specify how to manage it after new prices, fills or assignment notices arrive. We need a policy that values the effect of today's instruction on tomorrow's inventory and cash.

21.1 Two trading dates before the general recursion

Let s_1 contain tomorrow's market information, holdings and cash. At the last trading date define

$$V_1(s_1) = \max_{a_1 \in \mathcal{A}_1(s_1)} \mathbb{E}^{\mathbb{P}}[U(W_2^{\text{liq}}) - \psi_1(s_1, a_1) \mid s_1, a_1].$$

Here U is the chosen utility of final liquidation wealth, and ψ_1 is a noncash preference penalty. At the first date,

$$V_0(s_0) = \max_{a_0 \in \mathcal{A}_0(s_0)} \{-\psi_0(s_0, a_0) + \mathbb{E}^{\mathbb{P}}[V_1(s_1) \mid s_0, a_0]\}.$$

The first action can consume cash or create stock delivery, changing which actions are feasible tomorrow. Repeating a static spread optimizer ignores that continuation value unless its objective includes it.

Key idea

If I act now, I receive X, move to a random future state, and then behave optimally from there. With terminal utility, X is the current noncash reward $-\psi_t$; actual trading costs are already inside the wealth transition.

21.2 The state and action spaces

Let

$$s_t = (t, S_t, b_t, q_t, h_t, X_t, \mathfrak{p}_t, \mathfrak{d}_t, \ell_t, \mathcal{R}_t), \tag{21.1}$$

where $\mathfrak{p}_t$ includes pending stock/cash obligations, unsettled trades and unconfirmed exercise exposure; $\mathfrak{d}_t$ is dividend/calendar information; and $\mathcal{R}_t$ contains remaining risk budget and any running wealth maximum. The belief b_t carries latent variance, physical/pricing parameters and regime/model uncertainty.

An action $a \in \mathcal{A}_t(s)$ specifies option and share order instructions and exercise quantities for owned contracts. Filled trades can be random given the instruction. Assignment counts for shorts are not actions. For short quantity $q_j < 0$, define an assignment count $A_{j,t} \in \{0, \dots, -q_j\}$; its removal is $e_j = -A_{j,t}$. A conditional kernel for assignment may depend on moneyness, remaining time value, dividend dates and observed notices. It need not factor independently across legs.

Define a measurable transition $s_{t+1} = \mathfrak{T}(s_t, a_t, \xi_{t+1})$, where ξ_{t+1} contains market moves, fills, dividends, assignments and new observations under the physical law. It applies the ordered cash maps (1.2) and (20.9), accrual, pending settlement and Bayes update. Feasibility constrains intermediate as well as final state. If cash becomes insufficient, a declared forced-liquidation/default transition determines remaining obligations and losses; finite reserves do not guarantee solvency under an unbounded forecast law.

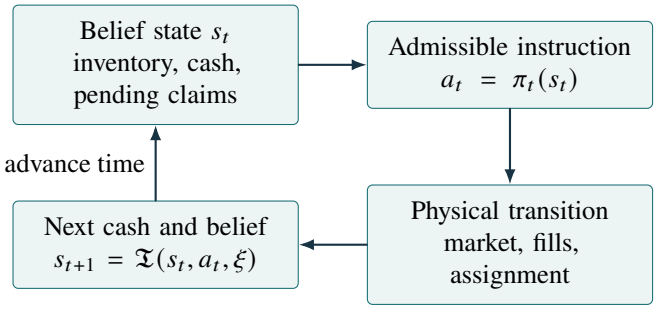


Figure 21.1: The generic state-action-transition loop. Assignment is part of the physical transition; only owned-option exercise is controlled.

21.3 Bellman recursion and terminal delivery

For increasing concave utility U, running noncash risk penalty $\psi_t \geq 0$, and final liquidation wealth W_T^{liq}, define

$$V_t(s) = \sup_{\pi \in \Pi_{\mathrm{adm}}} \mathbb{E}^{\mathbb{P}} \left[U(W_T^{\mathrm{liq}}) - \sum_{u=t}^{T-1} \psi_u(s_u, a_u) \mid s_t = s \right]. \tag{21.2}$$

Trading costs and financing are already in W_T^{liq}. On the discrete decision grid, conditioning on one step gives

$$V_T(s) = U(W_T^{\mathrm{liq}}(s)), \qquad V_t(s) = \sup_{a \in \mathcal{A}_t(s)} \left\{ -\psi_t(s, a) + \int V_{t+1}(s') K_t^P(\mathrm{d}s' \mid s, a) \right\}. \tag{21.3}$$

The derivation is the tower property followed by selection of the best feasible continuation; measurable maximizers require finite actions or compactness/continuity and integrable domination. A belief-state transition supplies the Markov property under the model. Ignoring pending notices or rough-volatility history can destroy it.

At terminal date, exercise or expiry removes each option according to its actual outcome, physical obligations enter stock/cash accounts, and remaining stock and live later-maturity options are liquidated or valued under a separately specified continuation contract. If the research horizon ends before all expiries, terminal marks are not terminal payoffs. Terminal liquidation costs and pending delivery funding must be included.

21.4 HJB, impulse control and stopping

A short-time Taylor expansion of the Bellman equation produces a differential equation for smooth diffusion control. Discrete option trades instead change the state by a jump. Their intervention operator compares the value after paying for a trade with the value of waiting.

For a continuously controlled diffusion state with generator $\mathcal{L}^{P,a}$, a smooth value and flow penalty ψ, expand the one-step Bellman equation over $\mathrm{d}t$, use Itô and divide by $\mathrm{d}t$:

$$0 = V_t + \sup_{a \in \mathcal{A}(s)} \{\mathcal{L}^{P,a} V - \psi(s,a)\}. \tag{21.4}$$

This Hamilton–Jacobi–Bellman equation requires regularity for a classical solution. When smoothness fails, a viscosity formulation or the original probabilistic control problem is needed [7]. Integer option trades are not infinitesimal diffusion controls.

For discrete position jumps with a fixed transaction cost, define the intervention operator $\mathcal{M}V(s) = \sup_{a \in \mathcal{A}^{\mathrm{jump}}(s)} V(\mathfrak{T}_{\mathrm{trade}}(s,a))$, where cash cost is deducted inside the transition. Then a schematic impulse-control inequality is

$$\max\{V_t + \mathcal{L}^P V - \psi,\ \mathcal{M}V - V\} = 0. \tag{21.5}$$

The first branch means waiting; the second means trade now. Holder exercise contributes an additional intervention/stopping branch with map (20.9). An American *market price* solves a $\mathbb{Q}$-stopping problem; the investor's exercise/trade decision solves a $\mathbb{P}$-utility problem with costs and portfolio interactions. Confusing the two would give the holder control over market pricing.

For assignment intensity $\lambda_j(s)$ and state map $\mathfrak{T}_j^{\mathrm{asg}}$, a Markov jump approximation adds $\lambda_j(s)[V(\mathfrak{T}_j^{\mathrm{asg}}(s)) - V(s)]$ to the physical generator. A count distribution or notice-delay belief is needed when more than one contract can be assigned or the event is not immediately observed. Dividends create scheduled jump conditions and alter stock entitlements.

21.5 Receding horizon and stochastic actions

Receding-horizon control optimizes over a limited future horizon, uses a declared terminal approximation and executes the first instruction. It is an approximation to (21.3). Scenario actions must agree whenever their observation histories agree; knowing a future assignment before submitting today's order is inadmissible.

A deterministic hedge rule maps s to one action. A probabilistic rule assigns a distribution $\pi(\mathrm{d}a \mid s)$. Given continuation value $Q(s,a)$, $\int Q(s,a)\pi(\mathrm{d}a \mid s) \le \sup_a Q(s,a)$; uncertainty about volatility does not by itself create a benefit from randomizing hedges. Bayesian integration, worst-case protection and smaller exposure all operate within deterministic decision rules.

If action entropy is an explicit preference on a finite feasible set, the maximization $\sum_a p_a Q_a - \tau \sum_a p_a \log p_a$ gives $p_a \propto e^{Q_a/\tau}$ by a normalization multiplier. As $\tau \downarrow 0$, probability concentrates on maximizers. This changes the objective and can deliberately incur lower expected financial utility. For continuous actions use relative entropy to a stated reference distribution, because differential entropy changes with action units.

Mathematical checkpoint

Would learning about a latent volatility parameter justify paying to trade, or can the same information be obtained while holding no position? When does an action actually change the information arriving next?

What we gained

- A two-date utility problem generalized to a belief-state policy.
- Separate controlled trades, owned exercise and exogenous assignment.
- Continuous, impulse and receding-horizon formulations with stated qualifications.

Where this goes next

The policy determines events along a realized path. We now telescope their cash identities into the exact total P&L and distinguish that total from a Greek attribution.

Chapter 22

Exact option P&L

What problem are we solving?

A detailed list of delta, gamma and vega contributions can still fail to balance. We first derive equity changes from event accounting, then use Greeks only to explain a component of that exact total.

22.1 Prove the ledger on one event at a time

For unchanged holdings, repricing changes equity by holdings times mark changes. A trade exchanges cash for securities, reducing equity only by its cost. Exercise exchanges an option for stock and strike cash. Settlement moves a pending obligation into the settled account. Each is an equality before and after a single event; summing equalities must preserve cash conservation.

Event ordering matters because a position earns only the price increments while it is actually held. We therefore mark live contracts before removing exercised contracts and count the removal separately.

22.2 An event-ordered identity

Partition time into ordinary market revaluation, trading, dividends, exercise recognition and settlement events. At each event apply market price changes first to incoming holdings, then trades/exercise at those marks. Let ΔC_u^{live} be the mark change of contracts while still alive, before any removal at u; a disappearing option is not also marked to zero in this term. Let $e_{j,u}$ be signed removals, and K_u, H_u realized trading and extra funding charges.

The exact marked-equity change over a finite event grid is

$$\begin{aligned} W_T - W_0 = & \sum_u q_{u-}^\top \Delta C_u^{\text{live}} + \sum_u h_{u-} \Delta S_u + \sum_u h_{u-} \Delta D_u \\ & + \sum_u I_u^{\text{cash}} - \sum_u (K_u + H_u) \\ & + \sum_{u,j} e_{j,u} \left[m_j \chi_j (S_u - K_j) - C_{j,u}^{\text{pre-removal}} \right]. \end{aligned} \tag{22.1}$$

Here I_u^{cash} is actual interest on the settled and pending cash accounts over the relevant interval. Stock dividend entitlements use the correct ownership/settlement convention; where the instantaneous model applies, $h_{u-} \Delta D_u$ is its expression. All quantities are in cash units.

To prove the identity, write $W = X + hS + q^\top C$ before and after each event. A market repricing changes wealth by $h\Delta S + q^\top \Delta C$. A dividend credits $h\Delta D$. A trade substitutes cash for marked securities and changes wealth only by $-K$, as in (1.2). Exercise substitutes delivery cash and shares for removed options, giving (20.10). Settlement transfers pending amounts into settled accounts without changing their total marked value. Summing these equalities telescopes to (22.1).

This proves self-financing path by path, without assuming a volatility model. Models determine the law of the terms, not the accounting signs.

22.3 Premium, mark and delivery

For one newly opened fixed portfolio with no intermediate exercise, initial premium outflow is $-q^\top C_0^{\text{entry}}$, final proceeds are $q^\top C_T^{\text{exit}}$, and cash accrual/financing transforms these into the financed gain. Mark-to-market option P&L $q^\top(C_T - C_0)$ already includes the change from the initial premium valuation; adding the initial premium again would double count it.

If a long call exercises, the removed claim is replaced by m shares and a cash payment mK; for a short call the deliveries reverse. Subsequent share moves belong to stock P&L. The terminal cash payoff $(S_T - K)_+$ is a convenient value representation of immediate delivery and liquidation, not evidence that the actual contract is cash settled. A put obeys the corresponding sign map.

An expired out-of-the-money contract disappears with zero mark in the ideal rational convention. Actual exercise contrary to that convention is represented by the observed removal count and its intrinsic transfer in (22.1). The policy must reserve for plausible assignment outcomes near expiry rather than infer certainty from an earlier mark.

22.4 Greek attribution refines, but does not add to, option P&L

If a price depends on two moving coordinates, changing one first and then the other gives an exact but order-dependent explanation. The double-integral identity below isolates the interaction term responsible for that dependence.

Within a smooth interval, multiply (10.6) by contract quantities and multipliers. The option mark term can be partitioned into delta, gamma, surface level/skew/maturity changes, theta, cross terms and remainder. These are subcomponents of $q^\top \Delta C$, not extra returns to add to (22.1).

There is no unique allocation of cross effects. For two state coordinates x, y, one exact sequential attribution is

$$\Delta_x V = V(x_1, y_0) - V(x_0, y_0), \tag{22.2}$$

$$\Delta_y V = V(x_1, y_1) - V(x_1, y_0). \tag{22.3}$$

Reversing the order gives different contributions. Their difference is

$$V(x_1, y_1) - V(x_1, y_0) - V(x_0, y_1) + V(x_0, y_0) = \int_{x_0}^{x_1} \int_{y_0}^{y_1} V_{xy}(u, v)\, \mathrm{d}v\, \mathrm{d}u, \tag{22.4}$$

under C^2 regularity. Averaging the two orders shares this interaction equally, but that is a convention. With more coordinates one may average over permutations or report an unallocated interaction/remainder term. Full gain remains invariant.

Financing, dividend and exercise terms also interact with the chosen mark convention. The audit record must specify event ordering, state coordinates, marking source, the treatment of cash rates, and whether exercise/delivery is immediate or pending. Without those choices, a detailed-looking P&L decomposition can fail to balance.

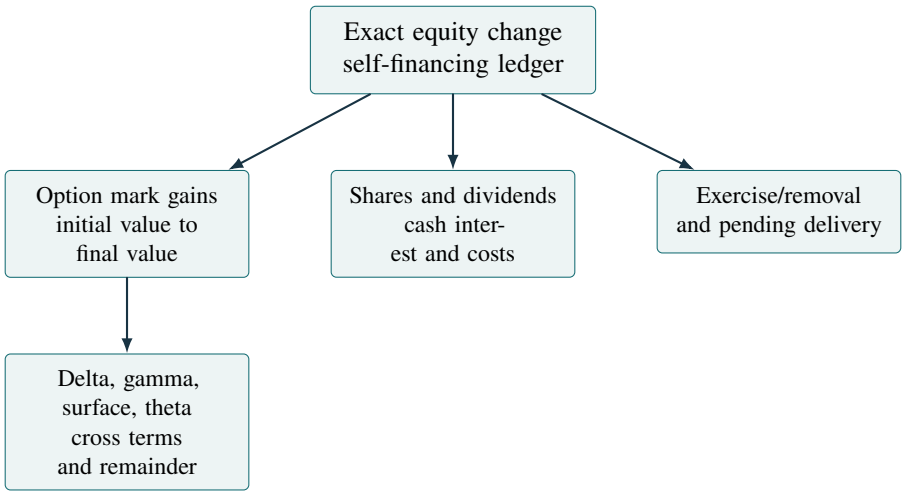


Figure 22.1: Attribution has levels. Greek terms explain the option-mark branch and must not be added a second time to the ledger.

22.5 Risk is a law of the full path

Variance of a local Greek approximation is a diagnostic. The relevant loss distribution includes jumps, costs, exercise and funding. Let $L^M = -\Pi^M$. Its physical expected shortfall, $\mathrm{ES}_\alpha(L^M \mid \mathcal{G}_t)$, a stress maximum over assignment patterns, and a margin-liquidity constraint protect different events. A terminal bounded payoff need not bound intermediate financing requirements.

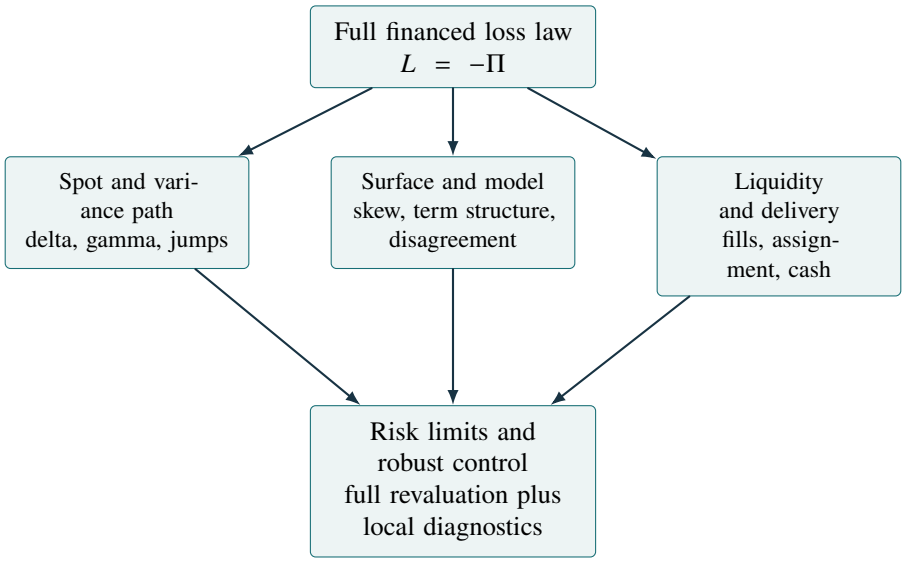


Figure 22.2: Nonlinear market risk, model risk and delivery funding risk enter the decision through distinct constraints or loss terms.

What we gained

- An event-ordered identity for total wealth change.
- Premium, live marks and physical removal counted exactly once.

- Greek attribution and path-dependent funding risk placed within that ledger.

Where this goes next

The ledger provides observable outcomes. The next chapter tests whether physical forecasting improved them, without confusing successful calibration with a successful trade.

Chapter 23

Statistical validation of the options system

What problem are we solving?

A model can match current prices, predict variance well, and still produce poor spread decisions after costs. We state these as separate hypotheses and evaluate each using observations appropriate to its claim.

23.1 Fit, forecast and decision

A quote residual compares a model with today's price. A forecast error compares a past conditional prediction with a later observation. A policy gain compares the full ledger with a matched benchmark. Keeping these comparisons separate prevents a good pricing calibration from being presented as evidence of alpha.

23.2 The claim to be investigated

At time t, the market supplies option bid and ask prices. Their relationship to no-arbitrage values constrains a family of pricing laws. Historical spot and option observations supply evidence about the physical law of tomorrow's market state. We ask whether a predeclared physical forecast improves the expected utility of a feasible, financed, cost-adjusted options policy relative to a benchmark using no such conditional information.

Let $\mathcal{G}_t^0 \subseteq \mathcal{G}_t$ be benchmark information and $\mathcal{G}_t^1$ add the proposed volatility or surface state. For future state path $Y_{t:t+h}$,

$$H_0^{\text{pred}} : \quad \mathcal{L}_{\mathbb{P}}(Y_{t:t+h} \mid \mathcal{G}_t^1) = \mathcal{L}_{\mathbb{P}}(Y_{t:t+h} \mid \mathcal{G}_t^0) \quad \mathbb{P}\text{-a.s.} \tag{23.1}$$

For a fixed policy construction π and benchmark π^0, the economic null is

$$H_0^{\text{econ}} : \quad \mathbb{E}^{\mathbb{P}}[\Pi_{t,t+h}^{\pi} - \Pi_{t,t+h}^{\pi^0}] \leq \delta_{\text{econ}}, \tag{23.2}$$

with a symbolic economically meaningful threshold $\delta_{\text{econ}} \geq 0$, matched capital and risk constraints, and all gains after costs. The alternative is strict inequality in the favorable direction. A positive option-selling mean may be risk compensation rather than information-based relative value.

23.3 What is observed and what is tested

The observed evaluation object is the realized ledger gain under a locked policy and a matched benchmark, with actual eligible quotes, costs and contract events. A latent parameter such as Hurst exponent is not directly a realized profit statistic. Let $D_t = \Pi^{\pi}_{t,t+h} - \Pi^{\pi^0}_{t,t+h}$ be gain improvement, and let Y_{t+h} be a predeclared observable forecast target such as a future American option price vector, a variance proxy or a surface-coordinate change.

A conditional forecast is $\widehat{F}_t = \mathcal{L}_{\mathbb{P}}(Y_{t+h} \mid \mathcal{G}_t)$. Its mean error is $u_{t+h} = Y_{t+h} - \mathbb{E}_{\widehat{F}_t} Y$. Correct conditional means require $\mathbb{E}[h_t^{\top} u_{t+h}] = 0$ for preselected $\mathcal{G}_t$-measurable instruments h_t. Distributional forecasts are compared by proper scores; for densities, expected negative log-score excess over the true density is a nonnegative Kullback–Leibler divergence. A quote-calibration residual tests today's pricing fit, not this conditional physical restriction.

23.4 Chronology and overlapping horizons

Partition observation dates into training, selection and final evaluation intervals. Specify the option eligibility rule, maturity/strike support, pricing challengers, physical estimation procedure, dividend treatment, costs, Greek tolerances and capital limits before final evaluation. At date t, a target ending at $t + h$ cannot enter training until it has been realized. Purge training targets whose payoff or label intervals overlap evaluation boundaries.

Different strikes on one date share spot and surface shocks. Different maturity trades can overlap for long periods, and daily hedging creates further dependence. The number of option quotes is therefore not the number of independent economic experiments. Aggregate contemporaneous strike-level results into the predeclared policy statistic, or use a covariance structure that accounts for within-date and temporal dependence.

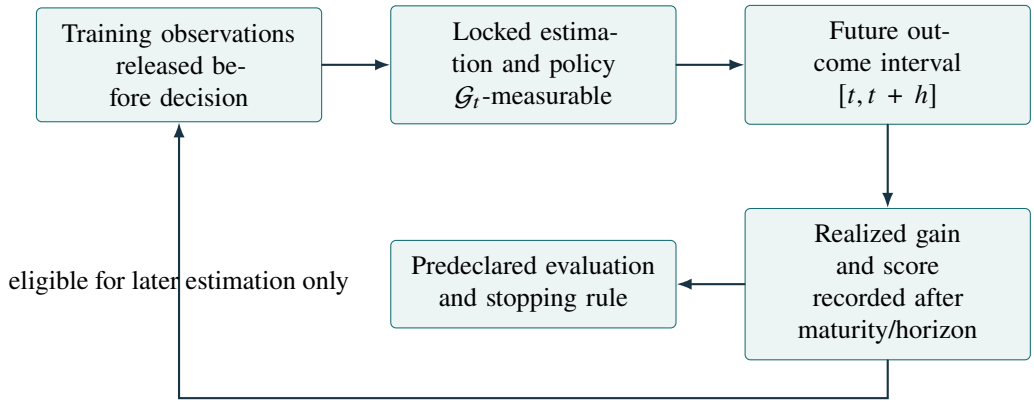


Figure 23.1: Information flow across an evaluation horizon. Outcomes become observations only after their realization.

23.5 A dependence-aware economic test

When gain horizons overlap, their shocks overlap. Expanding the variance of the sample mean introduces lag covariances, so the uncertainty cannot be computed from the number of quotes as if every quote were an independent experiment.

Under a stationary evaluation regime with suitable mixing, finite $2 + \epsilon$ moments and summable dependence, suppose $\sqrt{N}(\overline{D}_N - \mathbb{E}D_t) \Rightarrow \mathcal{N}(0, \omega^2)$, $\omega^2 = \sum_{\ell \in \mathbb{Z}} \mathrm{Cov}(D_t, D_{t-\ell}) > 0$. A

heteroskedasticity/autocorrelation-consistent estimate is

$$\widehat{\omega}^2 = \widehat{\gamma}_0 + 2\sum_{\ell=1}^{L_N}\left(1 - \frac{\ell}{L_N+1}\right)\widehat{\gamma}_\ell, \tag{23.3}$$

where $\widehat{\gamma}_\ell$ are centered sample lag covariances and bandwidth conditions must match the dependence assumptions [12]. Test $\mathbb{E}D_t \le \delta_{\text{econ}}$ with $\sqrt{N}(\overline{D}_N - \delta_{\text{econ}})/\widehat{\omega}$. The normal critical value is an asymptotic approximation under these assumptions; severe breaks or nonexistent moments invalidate it.

Risk compensation is addressed by matching the benchmark's declared risk exposures or by evaluating conditional utility differences under the same constraints. A regression on chosen risk factors is another benchmark model and requires its own specification; unexplained intercept is not automatically an arbitrage. Expected return, expected utility and pricing error remain separate test targets.

23.6 Multiplicity and the right to stop believing

Let M_{test} include all tested strike/maturity supports, model variants and policy thresholds, not merely the winner. Bonferroni with level α/M_{test} follows from the union bound if marginal tests are valid. Joint maximum-performance tests can be more powerful but must include the selection procedure [16]. Reusing the final evaluation set to choose another spread destroys its role as an independent confirmation.

For repeated monitoring dates, choose an error-spending sequence α_j with $\sum_j \alpha_j \le \alpha$, or a justified sequential procedure. The policy is suspended for present infeasibility, unresolved contract terms, a missing admissible pricing model, or an assignment-funding stress breach. Reject a predictive model when its predeclared conditional calibration test rejects. Reject a claimed economically meaningful edge when its upper confidence bound lies below δ_{econ}; lack of a positive lower bound alone is insufficient evidence, not proof of zero edge.

Compare Heston and rough alternatives on the same observable targets and eligible contracts. Short-time skew fit, variance-memory prediction, American exercise accuracy and after-cost hedge loss are different criteria. A model can win one and lose another. Report uncertainty in differences rather than create a universal numerical ranking. No empirical conclusion is supplied by this symbolic construction.

23.7 Literature choices and maintained system

The literature can now be read by mathematical role: replication, surface restrictions, volatility dynamics, control and testing. Model sophistication is assessed through the specific forecasts and decisions each tool is intended to improve.

Replication in Black and Scholes [2], Merton [11] is our simplest pricing laboratory. Density recovery from European prices [4], local volatility [6], and surface parameterizations [8] clarify what today's prices can identify. Heston's stochastic-variance model [10] provides a finite-dimensional comparator. Rough-volatility work [1, 9] motivates a Volterra alternative with different regularity and conditional-memory restrictions. None of these sources establishes profitability of the SPY policy defined here.

The main construction uses a joint physical belief over spot, latent variance and admissible future surface coordinates, together with a set of American pricing models consistent with current bid–ask intervals. Heston supplies a tractable baseline; a rough kernel model and local volatility are

challengers. Robust spread selection expresses a forecast only if its after-cost conclusion survives the declared model set. The control formulation follows the dynamic-programming foundations of Fleming and Soner [7]; convex subproblems use Boyd and Vandenberghe [3], while integer contracts and exercise decisions need additional treatment.

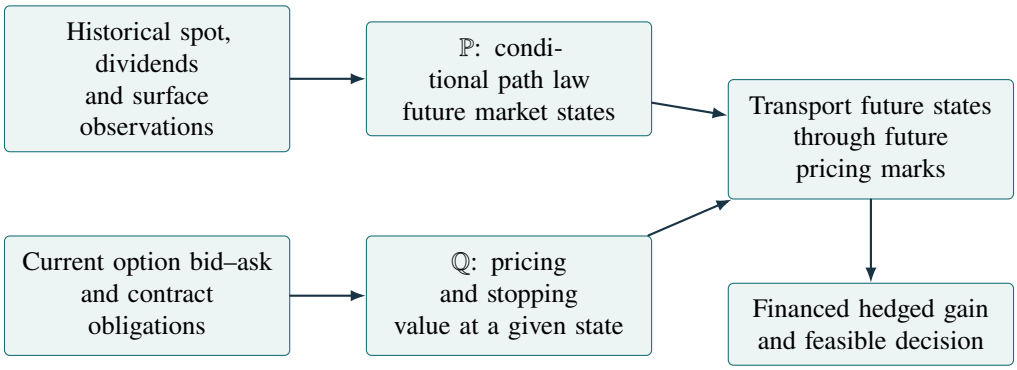


Figure 23.2: Physical forecasts and risk-neutral values are different maps. A future pricing state must itself be forecast under $\mathbb{P}$.

Mathematical checkpoint

If two pricing models agree on every quoted option today, what additional observations would identify the future surface dynamics needed to trade a spread through time?

What we gained

- Separate pricing-fit, predictive and economic test targets.
- Chronology and dependence handled at the level of policy outcomes.
- Model comparison, selection corrections and declared rejection rules.

Where this goes next

The appendices retain the technical qualifications and a full-horizon baseline. The closing two-page sheet keeps the one-period controller and adds a slow thesis state, physical paths with conditional pricing, instrument changes, multi-period control and separate execution.

Appendix A

Probability and no-arbitrage qualifications

What problem are we solving?

The main text introduced probability through finite outcomes and Brownian increments. These foundations specify the general information spaces and the conditions needed to pass from formal local calculations to valid pricing expectations.

A.1 Full information and observable information

Work on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ with the usual conditions. The subfiltration $\mathcal{G}_t \subseteq \mathcal{F}_t$ contains only released spot/option quotes, announced dividend information, funding and margin terms, and received execution/assignment notices. Positions chosen at t are $\mathcal{G}_t$-measurable; continuous-time holdings are predictable. Latent variance may be adapted to $\mathcal{F}_t$ without being observed in $\mathcal{G}_t$.

A.2 Continuous-time no arbitrage

Continuous-time results require stronger conditions. For locally bounded semimartingale discounted prices, no free lunch with vanishing risk is equivalent to existence of an equivalent local martingale measure, with admissibility appropriately defined. Turning a local martingale into a pricing expectation requires a true-martingale or uniform-integrability condition. This monograph uses selected pricing models for which those conditions are explicitly maintained; it does not infer them from a calibration [14].

A.3 Stopping integrability

Require the discounted reward process to be of class D, meaning its stopped values form a uniformly integrable family, with regularity sufficient for the stated stopping results.

A.4 Conditional laws and integrability

State spaces used here are standard Borel spaces, such as Euclidean spaces, finite model sets and separable spaces of continuous paths. Regular conditional laws can then represent state beliefs and

transition kernels. Bayes' formulas with densities are statements on observations with positive finite predictive density; conditioning on null events requires a specified version or limit.

A positive local martingale is a supermartingale and may lose expectation. To replace a replicated discounted value by an expectation, the stopped processes must converge with sufficient integrability. A class-D process has a uniformly integrable family of stopped values. This is why stopping and density-change chapters state true-martingale or class-D conditions, rather than conclude them from a zero drift alone.

What we gained

- Full and observed filtrations distinguished.
- The continuous-time no-arbitrage theorem stated with its scope.
- Conditional-law and integrability qualifications for pricing and filtering.

Where this goes next

The next appendix checks the stochastic-calculus identities used in pricing and measure change.

Appendix B

Calculus, pricing and measure-change checks

What problem are we solving?

A sign error in a measure change changes the model. These calculations verify the compensator, discounted gains and the equivalence between the smooth PDE and expectation formulations.

B.1 The exponential density compensator

Let $L_t = -\int_0^t \lambda_u^\top \, \mathrm{d}W_u^P$, so $\mathrm{d}[L]_t = \|\lambda_t\|^2 \, \mathrm{d}t$. For $Z_t = \exp(L_t - \frac{1}{2}[L]_t)$, Itô's formula gives

$$\mathrm{d}Z_t = Z_t \left(\mathrm{d}L_t - \tfrac{1}{2} \, \mathrm{d}[L]_t \right) + \tfrac{1}{2} Z_t \, \mathrm{d}[L]_t = -Z_t \lambda_t^\top \, \mathrm{d}W_t^P. \tag{B.1}$$

This establishes a positive local martingale. The true-martingale condition such as Novikov is the extra step required to make $\mathbb{E}^{\mathbb{P}} Z_T = 1$ and define a probability. A positive local martingale can lose expectation; omitting this distinction can turn a formal drift change into an invalid pricing argument.

B.2 Discounted gains

For a no-dividend diffusion S, differentiation of S/B gives $\mathrm{d}(S/B) = B^{-1}(\,\mathrm{d}S - rS\,\mathrm{d}t)$. With dividend process D, add $B^{-1}\,\mathrm{d}D$. Thus the pricing drift restriction is that the finite-variation part of $\mathrm{d}S + \mathrm{d}D - rS\,\mathrm{d}t$ vanish under $\mathbb{Q}$. This proves the use of $r - \delta$ for an ex-dividend stock with continuous yield and explains the financed hedge integrand in (19.5).

B.3 Verification of the pricing PDE

For a smooth solution of (4.5), apply Itô to $B_t^{-1} c(t, S_t)$ under $\mathbb{Q}$. Its drift vanishes, leaving $B_t^{-1} \sigma S_t c_s \, \mathrm{d}W_t^Q$. If this stochastic integral is a true martingale and the terminal condition is integrable, conditioning at t gives the pricing expectation. Conversely, regularity of the conditional expectation yields the PDE. Smoothness and integrability are the hypotheses linking the probabilistic and differential formulations, not automatic consequences of writing a formula.

What we gained

- The exponential-density drift cancellation.
- The dividend-adjusted discounted-gain equation.
- A PDE verification with its true-martingale requirement.

Where this goes next

Surface inversion adds another layer of differentiation; the next appendix carries out its conditioning and curvature calculations.

Appendix C

Surface calculus and inverse conditioning

What problem are we solving?

Implied variance is a nonlinear coordinate for price. To test strike convexity, its slope and curvature must both be carried through the chain rule.

C.1 Technical note: SSVI admissibility restrictions

With smooth nondecreasing $\theta(\tau)$, the sufficient static-arbitrage restrictions of Gatheral and Jacquier [8] include, for all relevant θ,

$$0 \le \frac{\mathrm{d}}{\mathrm{d}\theta}[\theta\phi(\theta)] \le \frac{1+\sqrt{1-\rho_s^2}}{\rho_s^2}\phi(\theta), \tag{C.1}$$

$$\theta\phi(\theta)(1+|\rho_s|) < 4, \qquad \theta\phi(\theta)^2(1+|\rho_s|) \le 4. \tag{C.2}$$

For $\rho_s = 0$, the derivative upper bound is interpreted as infinite. Require $\theta(\tau) \to 0$ at zero maturity with an appropriate vanishing off-the-money limit, and the usual call boundary conditions. These are sufficient restrictions for this specific family, not a universal characterization of every arbitrage-free surface.

For a twice differentiable positive smile $w(k)$, strike convexity can be checked through

$$\mathfrak{g}(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \tag{C.3}$$

The chain-rule calculation in Appendix C gives $c^E_{KK} = D(t,T)\varphi(d_2)\mathfrak{g}(k)/(K\sqrt{w})$. Nonnegative $\mathfrak{g}$, together with valid tail limits, ensures nonnegative density. Enforcing isolated quote fits without these conditions can create impossible prices between quotes.

C.2 Strike curvature in total-variance coordinates

For fixed maturity, put $K = Fe^k$, $w = w(k) > 0$, and define normalized call $c(k) = \Phi(d_+) - e^k\Phi(d_-)$, where $d_\pm = -k/\sqrt{w} \pm \sqrt{w}/2$. Use $\varphi(d_+) = e^k\varphi(d_-)$. Then

$$c_k = -e^k\Phi(d_-) + e^k\varphi(d_-)\frac{w'}{2\sqrt{w}}. \tag{C.4}$$

Let $a = w'/(2\sqrt{w})$. Since

$$d'_- = -\frac{1}{\sqrt{w}} + \frac{kw'}{2w^{3/2}} - \frac{w'}{4\sqrt{w}}, \quad a' = \frac{w''}{2\sqrt{w}} - \frac{(w')^2}{4w^{3/2}}, \tag{C.5}$$

differentiate again and subtract c_k:

$$c_{kk} - c_k = e^k \varphi(d_-)\{a' - d'_-(1 + d_- a)\}. \tag{C.6}$$

Now $1 + d_- a = 1 - kw'/(2w) - (w'/4)$, and $-\sqrt{w}\, d'_- = 1 - kw'/(2w) + (w'/4)$. Their product is $(1 - kw'/(2w))^2 - (w')^2/16$. Adding $\sqrt{w}\, a' = w''/2 - (w')^2/(4w)$ yields $\mathfrak{g}(k)$ in (C.3). Because $\partial_K = K^{-1}\partial_k$, restoring the factor DF gives $c^E_{KK} = D\varphi(d_-)\mathfrak{g}(k)/(K\sqrt{w})$. This derivation explains why smile slope and curvature both enter density positivity.

C.3 What a noisy inverse amplifies

For a calibration map $F : \Theta \to \mathbb{R}^m$, local perturbations satisfy $\delta C \approx J_F \delta\vartheta$. An SVD $J_F = UDV^\top$ shows that recovering the i-th parameter direction divides the price perturbation by singular value D_{ii}. A singular direction is locally unidentified; a nearly singular one is noise-sensitive. This applies to implied volatility, Heston calibration and rough-kernel parameters, not only to matrix inverses in portfolio optimization.

What we gained

- The total-variance density factor derived step by step.
- A strike-coordinate chain rule with the discount factor restored.
- An SVD explanation of calibration sensitivity.

Where this goes next

The next appendix develops the variance moments and Gaussian regularity details behind the volatility models.

Appendix D

Variance moments, roughness and memory

What problem are we solving?

The main text emphasized the state information needed by each model. Here we derive the stationary variance density, specify the fractional Gaussian comparison, and verify the information encoded by a forward curve.

D.1 The square-root stationary density

For variance drift $b(v) = \kappa(\theta - v)$ and squared diffusion $a(v) = \xi^2 v$, a stationary density $p(v)$ with zero boundary probability flux satisfies $b(v)p(v) - \frac{1}{2}(a(v)p(v))' = 0$. Thus

$$\frac{p'(v)}{p(v)} = \frac{2\kappa\theta/\xi^2 - 1}{v} - \frac{2\kappa}{\xi^2}. \tag{D.1}$$

Integrating gives $p(v) = Cv^{\alpha-1}e^{-\beta v}$, $\alpha = 2\kappa\theta/\xi^2 > 0, \beta = 2\kappa/\xi^2 > 0$. Normalization via the gamma integral gives $C = \beta^\alpha/\Gamma(\alpha)$. The same integral yields mean $\alpha/\beta = \theta$ and variance $\alpha/\beta^2 = \theta\xi^2/(2\kappa)$.

To verify the conditional variance formula without omitting the second-moment step, let $s(h) = \mathbb{E}[v_{t+h}^2 \mid v_t = v]$ and $m(h)$ be the mean. Then

$$s(h) = e^{-2\kappa h}v^2 + (2\kappa\theta + \xi^2)\int_0^h e^{-2\kappa(h-u)}[\theta + (v-\theta)e^{-\kappa u}]\,\mathrm{d}u.$$

The two integrals are $\theta(1-e^{-2\kappa h})/(2\kappa)$ and $(v-\theta)(e^{-\kappa h} - e^{-2\kappa h})/\kappa$. Subtracting $[\theta + (v-\theta)e^{-\kappa h}]^2$, grouping the coefficients of v and θ, gives (13.3).

D.2 Fractional Brownian motion and regularity

Fractional Brownian motion B_t^H, $H \in (0,1)$, is the centered Gaussian process with covariance

$$\mathbb{E}[B_t^H B_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H}). \tag{D.2}$$

Gaussianity and this covariance determine all finite-dimensional distributions. Subtracting covariance terms gives $\mathbb{E}[(B_{t+h}^H - B_t^H)^2] = |h|^{2H}$; increments are stationary and their standard deviation scales as

$|h|^H$. Gaussian moments give $\mathbb{E}|\Delta B^H|^p = c_p|h|^{pH}$, with c_p the absolute p-th moment of a standard normal. The continuity criterion based on these moment bounds yields sample paths locally Hölder of every order below H, but not of order above H.

At $H = 1/2$, covariance becomes $\min(s,t)$, the Brownian case. For $H < 1/2$, increments shrink more slowly as the time step shrinks, so paths are less regular. Adjacent increments of equal length h have covariance $\frac{1}{2}[(2h)^{2H} - 2h^{2H}] < 0$. Thus local roughness is not synonymous with positively correlated long memory. A slowly decaying covariance at large lags is a separate property.

For $H \neq 1/2$, fractional Brownian motion is not a semimartingale in its natural filtration. An unqualified Itô integral against it is therefore not available. The trading model in Chapter 17 avoids making the traded spot itself a fractional-Brownian diffusion; spot remains a semimartingale driven by ordinary Brownian motion. Roughness appears in a nontraded volatility state. This distinction is central to the use of Bayer et al. [1], Gatheral et al. [9].

D.3 Short-maturity skew: a restricted comparison

The following scaling is a perturbative explanation, not a universal asymptotic theorem. Near an initially flat forward variance v_0, expand $v_u \approx v_0(1 + \eta Y_u)$ to first order in η. The covariance between a leading log-return Brownian shock and integrated variance involves

$$\int_0^\tau \int_0^u \mathcal{K}_H(u-s)\,\mathrm{d}s\,\mathrm{d}u = \frac{\sqrt{2H}\,\tau^{H+3/2}}{(H+1/2)(H+3/2)}. \tag{D.3}$$

Brownian leverage correlation contributes a factor ρ. After normalizing a return third moment by its $\tau^{3/2}$ variance scale, standardized skew is of order $\rho\eta\tau^H$. In a small-skew European implied-volatility expansion, the local strike slope introduces an additional $\tau^{-1/2}$, giving the characteristic order $\partial_k \sigma_{\mathrm{imp}}(0,\tau) = O(\rho\eta\tau^{H-1/2})$. For $H = 1/2$, this order is bounded; for $H < 1/2$, it permits increasing slope magnitude at short horizons.

Interchanging small-vol-of-vol and small-maturity limits needs proof in a specific model. Jumps, nonflat initial curves and exercise near dividend dates can change the observation. The policy therefore tests future surface and hedged-gain predictions directly instead of treating a short-time scaling as universal evidence for rough volatility.

D.4 The forward-variance curve is sufficient for the Gaussian driver

For future $u > t$, the conditional mean of Y_u is $A_t(u)$ and its conditional covariance is the future-kernel integral in Chapter 17. These quantities determine a conditional Gaussian process. If $\xi_0(u) > 0$ and $\eta \neq 0$, (17.8) can be inverted pointwise:

$$A_t(u) = \eta^{-1} \log \frac{\xi_t(u)}{\xi_0(u)} + \frac{\eta}{2}\int_0^t \mathcal{K}_H(u-s)^2\,\mathrm{d}s. \tag{D.4}$$

Hence the entire curve $\xi_t(\cdot)$, with known parameters and initial curve, encodes the relevant Gaussian-driver past for future variance. A finite collection of option quotes does not observe this entire curve without an inverse-model assumption.

What we gained

- The CIR stationary density and conditional variance calculation.

- Fractional Brownian regularity distinguished from the Volterra driver.
- The forward curve's sufficiency for the modeled Gaussian past.

Where this goes next

The next appendix checks the price derivatives and weighted hedge equations used in portfolio geometry.

Appendix E

Greek derivatives and constrained exposure

What problem are we solving?

The BSM sensitivities simplify through a cancellation between normal density terms. We verify it before deriving a weighted minimum-size hedge.

E.1 The cancellation behind BSM Greeks

From the definitions, $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1^2 - d_2^2 = 2\log(s/K) + 2(r-\delta)\tau$. Exponentiating the normal density ratio proves

$$se^{-\delta\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2). \tag{E.1}$$

Differentiating (6.5) in s, the density terms cancel because $\partial_s d_1 = \partial_s d_2 = 1/(s\sigma\sqrt{\tau})$, leaving $\Delta = e^{-\delta\tau}\Phi(d_1)$. Another spot derivative gives gamma.

Write $d_1 = A/(\sigma\sqrt{\tau}) + \sigma\sqrt{\tau}/2$, with $A = \log(s/K) + (r-\delta)\tau$. Then $\partial_\sigma d_1 = -d_2/\sigma$ and $\partial_\sigma d_2 = -d_1/\sigma$. The price derivative cancels to $\nu = se^{-\delta\tau}\varphi(d_1)\sqrt{\tau}$. Differentiating delta in σ gives vanna, while differentiating vega and using $\varphi'(d_1) = -d_1\varphi(d_1)$ gives Volga $= \nu d_1 d_2/\sigma$. Theta follows either by differentiating in $-\tau$, including the discount factors, or by rearranging (4.5) with the derived delta and gamma.

E.2 Weighted minimum-cost neutrality

For $H_q \succ 0$, minimize $\frac{1}{2}q^\top H_q q$ subject to $Gq = b$, assuming G has full row rank and the equality is feasible. Stationarity gives $H_q q - G^\top\lambda = 0$, so

$$q^\star = H_q^{-1}G^\top(GH_q^{-1}G^\top)^{-1}b. \tag{E.2}$$

The second inverse exists because $G^\top z \neq 0$ for nonzero z. The matrix H_q can encode a quadratic risk or liquidity metric; choosing it changes the criterion being minimized. Bounds, integer quantities and assignment constraints can prevent this continuous solution from being admissible.

What we gained

- A shared normal-density identity behind the Greeks.

- The vanna and volga derivatives with consistent coordinates.
- A weighted constrained hedge and its invertibility conditions.

Where this goes next

The final appendix checks stopping signs and cash delivery independently of the pricing model.

Appendix F

Stopping and ledger verification

What problem are we solving?

Pricing, exercise and delivery meet at a contract event. The remaining checks ensure that stopping inequalities have the correct signs and that recognized obligations enter wealth once.

F.1 Why the obstacle has its signs

The holder can exercise, so $c \geq g$. The discounted American value is a supermartingale under a selected pricing model; therefore its continuation drift satisfies $c_t + \mathcal{L}^{Q} c - rc \leq 0$. When $c > g$, immediate exercise is suboptimal and the drift is zero under optimal continuation. These complementarity statements yield (18.3). The stopping rule $\tau^{\star} = \inf\{u \geq t : c^{A}(u, Z_u) = g(S_u)\} \wedge T$ is optimal under the regularity assumptions ensuring attainment; without them use the essential-supremum formulation and approximating stopping times.

F.2 Delivery conservation

For one signed removal e, subtract the pre-event equity from the post-event equity:

$$[X - m\chi K e] + [h + m\chi e]S + [q - e]C - (X + hS + qC) \tag{F.1}$$

$$= e[m\chi(S - K) - C]. \tag{F.2}$$

This calculation proves the assignment signs for both calls and puts. For immediate rational exercise $C = mg(S)$, the value transfer is zero at the exercise boundary. A pending deliverable has the same mark, then migrates into settled cash and stock. Counting both the pending claim and the delivered stock after settlement would duplicate wealth.

F.3 Terminal accounting and benchmark gain

Let W_T^{liq} be equity after all prescribed terminal deliveries and liquidation costs. The gain in excess of the money-market benchmark is

$$\Pi_{0,T} = W_T^{\mathrm{liq}} - \frac{B_T}{B_0} W_0. \tag{F.3}$$

This definition includes initial premiums, financing, hedge trades and delivery through the cash identity. If the research criterion compares two policies, apply the same terminal liquidation, dividend convention and initial capital to both.

With no positions, no costs and the baseline rate on all cash, $\Pi_{0,T} = 0$. At unchanged marks, buying an option reduces cash and increases marked option value by equal amounts. A short-option premium creates cash plus a liability. A dividend price drop of exactly the dividend amount is offset by the dividend credit for a qualifying share holder. These are symbolic conservation checks independent of the chosen volatility model.

F.4 One fully declared baseline within the model family

To make the general maps concrete, fix a research block, decision grid and eligible option set. The baseline physical state is (S, v) with the Heston equations of Chapter 13; no independent surface factor is introduced. Thus its forecast of surface motion is exactly the motion induced by spot and variance. Deterministic declared cash-dividend dates are applied as price jumps between diffusion intervals. Only announced dividend amounts enter the current information; uncertain later amounts belong to an explicitly chosen dividend law.

Fix a compact parameter domain for $\vartheta = (\mu, \kappa_P, \theta_P, \kappa_Q, \theta_Q, \xi, \rho)$ with positive mean-reversion/variance coefficients, $|\rho| < 1$, both Feller conditions, and the required finite-horizon moment and density-martingale conditions. Use the risk-price relation of Chapter 7 to connect the two drift specifications; exclude parameter pairs that fail it. A proper prior with density positive on this domain and an initial variance density complete the statistical model. Priors and bounds are symbolic design choices fixed before observations.

The physical transition density is the joint diffusion transition characterized by the stated generator and dividend maps. Its likelihood combines exact observed spot increments with a specified variance-proxy measurement

$$\log Y_t^v = \log v_t + \epsilon_t^v, \qquad \epsilon_t^v \sim \mathcal{N}(0, \sigma_{\text{obs}}^2), \quad \sigma_{\text{obs}} > 0,$$

where $Y_t^v > 0$ is a declared proxy with this maintained noise model. Conditional quote observations have a declared centered error density around American model values; bid–ask widths determine observation intervals or weights. Independence of the measurement errors conditional on state is the baseline assumption. Cross-sectional error dependence is a challenger. Bayes' recursion in Chapter 19, rather than replacement by observed squared return, defines the inferred variance.

This is an explicit generative hypothesis: the surface at a state is the vector of American stopping values under the paired $\mathbb{Q}$ law. A state with no acceptable quote fit rejects the baseline. The rough and local-volatility challengers retain their own path states and are compared through full gain laws. Adding a freely moving risk-price surface state requires specifying its joint $\mathbb{P}$ and $\mathbb{Q}$ transitions; it is not silently inserted into this baseline.

For assignment on the decision grid, one transparent baseline is

$$A_{j,t+1} \mid s_t, a_t \sim \text{Binomial}\left((-q_j^+)_+, \frac{1}{1 + \exp(-\beta_A^\top f_{j,t})}\right),$$

where $\beta_A \in \mathbb{R}^5$ is an unknown assignment parameter and $f_{j,t} \in \mathbb{R}^5$ contains a constant, log moneyness, remaining time, option time value and announced near-term dividend amount, each in fixed symbolic units. Conditional independence across contracts is an explicit baseline restriction; joint assignment stress cases relax it. The assignment likelihood updates β_A from received counts. A notice-delay law and pending-claim filter are included when notices are delayed. This assignment model is not the risk-neutral exercise boundary and must be tested separately.

Take bounded integer option instructions and bounded share-lot instructions, so the action set is finite; fills have a specified conditional kernel, with the immediately filled cost model of Chapter 20

as the simplest case. Impose the Greek tolerances and full assignment stress reserves in (19.8). Use the increasing concave, at-most-linear-growth terminal utility

$$U(w) = w - \lambda_U \sqrt{a_U^2 + w^2}, \qquad a_U > 0, \quad 0 < \lambda_U < 1. \tag{F.4}$$

Its derivative is $1 - \lambda_U w / \sqrt{a_U^2 + w^2} > 0$, and its second derivative is $-\lambda_U a_U^2 / (a_U^2 + w^2)^{3/2} < 0$. Finite first wealth moments therefore suffice for expected utility; an exponential utility would need stronger loss-tail integrability.

The final dynamic policy maximizes this terminal expected utility with declared running exposure penalties through (21.3). The mean-variance/expected-shortfall spread problem (19.7) is a useful static or receding-horizon approximation to that policy, not an assertion that these different criteria produce the same optimum. When robustness is included dynamically, use one-step conditional ambiguity sets stable under pasting, giving

$$V_t(s) = \max_{a \in \mathcal{A}_t(s)} \left\{ -\psi_t(s,a) + \inf_{K \in \mathfrak{K}_t(s,a)} \int V_{t+1}(s') K(\mathrm{d}s') \right\}.$$

Stability under pasting means that independently selected allowed conditional kernels at successive histories remain in the allowed path-law family. It makes this worst-case recursion time consistent. A fixed unknown model chosen once for the entire horizon instead belongs in the belief state; taking a new worst-case model at every date solves a different, more adversarial problem.

For fixed symbolic priors, bounds, error laws, cost maps, assignment coefficients and utility constants, this baseline defines every distribution, feasible action and cash transition. Nothing in the construction assigns those unknown constants empirical values or guarantees that the baseline survives its tests.

What we gained

- Complementarity conditions behind the exercise obstacle.
- Delivery conservation for calls and puts with either position sign.
- A benchmark gain that includes liquidation and pending obligations.

Where this goes next

Return to the closing operational sheet or use the synthesis to compare this nonlinear gain map with the futures settlement system.

Bibliography

[1] Christian Bayer, Peter Friz, and Jim Gatheral. Pricing under rough volatility. *Quantitative Finance*, 16(6):887–904, 2016. doi: 10.1080/14697688.2015.1099717.

[2] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

[3] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. URL https://web.stanford.edu/~boyd/cvxbook/.

[4] Douglas T. Breeden and Robert H. Litzenberger. Prices of state-contingent claims implicit in option prices. *The Journal of Business*, 51(4):621–651, 1978. doi: 10.1086/296025.

[5] Cboe Global Markets. Product comparison: Index options and SPY ETF options. Official contract comparison, n.d. URL https://www.cboe.com/tradable_products/product_comparison/. Accessed September 26, 2026.

[6] Bruno Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994. URL https://www.risk.net/derivatives/equity-derivatives/1500211/pricing-with-a-smile. Publisher retrospective identifies the original January 1994 article.

[7] Wendell H. Fleming and H. Mete Soner. *Controlled Markov Processes and Viscosity Solutions*. Springer, 2 edition, 2006. doi: 10.1007/0-387-31071-1.

[8] Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014. doi: 10.1080/14697688.2013.819986. URL https://arxiv.org/abs/1204.0646.

[9] Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018. doi: 10.1080/14697688.2017.1393551.

[10] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.

[11] Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973. doi: 10.2307/3003143.

[12] Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987. doi: 10.2307/1913610.

[13] R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *The Journal of Risk*, 2(3):21–41, 2000. doi: 10.21314/JOR.2000.038.

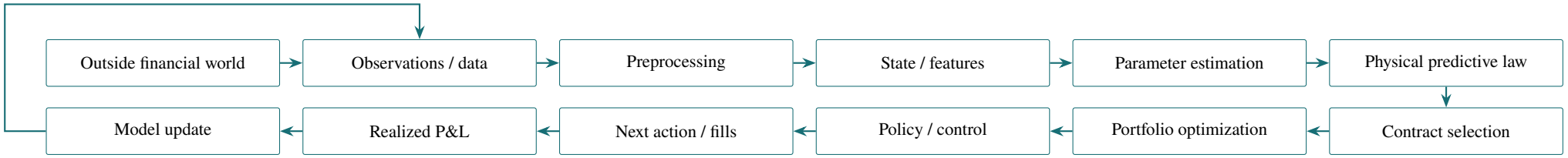
[14] Steven E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004. doi: 10.1007/978-1-4757-4296-1.

[15] The Options Clearing Corporation. Characteristics and risks of standardized options. June 2024 edition, 2024. URL https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document. Current edition checked September 26, 2026; especially Chapters III, VIII and X.

[16] Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152.

SPY options: thesis paths under P, option values under Q

Baseline: one-period Heston/utility MPC. Extension: a slow valuation anchor, a fixed economic deadline and adaptive expression.



Keep the physical thesis separate from its current instrument. Observations update beliefs; strategic decisions and child-order execution use separate clocks.

Three clocks. Observations at o_j may be intraday; strategic decisions at t_k may be daily; the economic horizon ends at a fixed T^{th}, e.g. four weeks. H is the number of planned decision intervals, with $t_{k+H} \le T^{\text{th}}$. Released information is $\mathcal{F}_k$. Freeze priors, proxy definitions, calendar/surprise laws, fits, bounds, simulation count P, seed, numerical grids and test blocks. Between decisions observe/filter; only scheduled decisions or mandatory risk/exercise events authorize portfolio changes.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O0. Released data / features $S, Y^v > 0$; J listed legs with strike K_j, expiry T_j, multiplier m_j, sign $\chi_j = \pm 1$. $C, \omega \in \mathbb{R}^J$, $X \in \mathbb{R}^{2+2J}$; economic proxy $y \in \mathbb{R}$.	Synchronize spot, bid/ask, dividends, funding, liquidity and notices; reject stale/crossed quotes. For an interval of Δ years set $Y^v = \max\{\epsilon_v, \sum_l (\Delta_l \log S)^2/\Delta\}$. Dollar midpoint $C_j = m_j(\text{bid}_j + \text{ask}_j)/2$; $\omega_j = \max\{\omega_{\min}, m_j(\text{ask}_j - \text{bid}_j)/2\}$, positive floors. $X = (S, \log Y^v, C, \omega)$, plus mask/calendar. Map released economic evidence to a predeclared log-valuation proxy y; record its formula and timestamps. **To O1–O4.**	Records; NumPy `(J,)`. Optional option grid `(N_K,N_T,2)` plus mask. Economic evidence is data, not a retrospectively revised story.
O1. Fast market / slow thesis $Z = (S, v, u) \in \mathbb{R}^2_+ \times \mathbb{R}$; u: latent log-valuation anchor; $d = u - \log S$: convergence gap. $c \in \mathbb{R}^P$: catalyst surprises. $\theta \in \Theta \subset \mathbb{R}^{12+P}$: seven Heston parameters plus slow coefficients.	Between decision dates, freeze u_k and use $\mathrm{d}S/S = \mu_k^P(S, u_k)\,\mathrm{d}t + \sqrt{v}\,\mathrm{d}W_1^P$, $\mathrm{d}v = \kappa_P(\bar{v}_P - v)\,\mathrm{d}t + \xi\sqrt{v}(\rho\,\mathrm{d}W_1^P + \sqrt{1-\rho^2}\,\mathrm{d}W_2^P)$, $\mu_k^P = \mu + \kappa_d(u_k - \log S)$. Slow model: $u_{k+1} = \bar{u} + \phi(u_k - \bar{u}) + \beta^\top c_{k+1} + \sigma_u \eta$, $y = u + \tau_y \nu$, $\beta \in \mathbb{R}^P$. Independent $\eta, \nu \sim \mathcal{N}(0,1)$; fit $0 < \phi \le 1$, $\sigma_u, \tau_y > 0$. At announced releases $c \sim \mathcal{N}(\bar{c}, C_c)$, fitted to past surprises; otherwise $c = 0$. Under $\mathbb{Q}$, stock drift is r, variance drift $\kappa_Q(\bar{v}_Q - v)$; ξ, ρ are shared. Risk prices: $\lambda_1 = (\mu_k^P - r)/\sqrt{v}$, $\xi\sqrt{v}(\rho\lambda_1 + \sqrt{1-\rho^2}\lambda_2) = \kappa_P(\bar{v}_P - v) - \kappa_Q(\bar{v}_Q - v)$. Impose $2\kappa_P\bar{v}_P, 2\kappa_Q\bar{v}_Q \ge \xi^2$, $\lvert\rho\rvert < 1$, positive variance coefficients and admissible moment/martingale bounds. Apply announced cash-dividend jumps on the stated positive-price domain. $\kappa_d = 0$ recovers constant drift. **To O2–O3.**	Fast `(2,)` / slow scalar; parameter vector `(12+p,)`. Conditional transitions are callables. Only $\mathbb{P}$ forecasts thesis convergence; the anchor is not a $\mathbb{Q}$ stock price.
O2. Filter / estimate $b_k \in \mathcal{P}(\mathbb{R}_+ \times \mathbb{R} \times \Theta)$: posterior of (v, u, θ). $\widehat{\theta}, \widehat{v}, \widehat{u}$: mode; ℓ_k: measurement likelihood.	Start with a fixed proper prior. Measurements: $\log Y^v \mid v \sim \mathcal{N}(\log v, \sigma_v^2)$, $y \mid u \sim \mathcal{N}(u, \tau_y^2)$, $C_j \mid Z, \theta \sim \mathcal{N}(c_j^A, \omega_j^2)$, conditionally independent; omit missing observations. With the O1 joint transition, $b_{k+1}(v,u,\theta) \propto \ell_{k+1}(v,u,\theta) \iint p_\theta^P(S_{k+1}, v, u \mid S_k, v', u') b_k(v', u', \theta)\,\mathrm{d}v'\,\mathrm{d}u'$. Normalize; use quadrature/particles and update released measurements between decisions. Parameters remain fixed along each future path; decisions integrate the posterior, not just its mode. Mode Greeks are a local constraint approximation. **To O3–O5.**	Weighted particles `(N_particles,14+p)` or quadrature; fit by likelihood/Bayes update. Observation noise $\sigma_v > 0$; retain posterior uncertainty.
O3. Path forecast then reprice P_k^H: physical law on $Z_{k:k+H}$. $c_j^A \in \mathbb{R}_+$: dollars/contract; $G \in \mathbb{R}^{3\times J}$: delta/gamma/vega. $g_j(S) = [\chi_j(S - K_j)]_+$.	$\mathsf{P}_k^H = \int \text{Law}_\theta^P(Z_{k:k+H} \mid S_k, v, u) b_k(\mathrm{d}v, \mathrm{d}u, \mathrm{d}\theta)$. $\mathbb{P}(Z_{k+H} \mid \mathcal{F}_k)$ is only its endpoint marginal. Draw real-world *paths*, catalyst outcomes and future noisy observations. At *each* simulated state $Z_l^{(p)}$, mark each live option with the paired $\mathbb{Q}$ Snell solver: $c_j^A(l, S, v) = \max\{m_j g_j(S), e^{-\int_l^{l'} r\,\mathrm{d}t} \mathbb{E}_\theta^Q[c_j^A(l', S', v') \mid S, v]\}$, terminal $m_j g_j$. Refine the exercise grid (including dividends/expiry) to a fixed tolerance. $G_{\cdot j} = (\partial_S c_j^A, \partial_{SS} c_j^A, 2\sqrt{v}\partial_v c_j^A)^\top$. Thus $\mathbb{P}$ weights future outcomes; $\mathbb{Q}$ gives conditional option marks, never probabilities of thesis success. Re-filter synthetic observations before each future action. **To O5–O7.**	Paths `(P,H+1,3)` (subdivide for diffusion/events); marks `(P,H+1,J)`, Greeks `(3,J)`; price/filter callables.
O4. Thesis mode / instruments $p_k = b_k(d_k > \delta)$; $z_k \in \{0, 1, \dagger\}$: waiting, active, closed; $p_{\text{on}} > p_{\text{off}}$. $E_k \in \{0,1\}^{m\times J}$: dated leg selector; chosen $C \in \mathbb{R}^m$, $G \in \mathbb{R}^{3\times m}$.	Activate at $p \ge p_{\text{on}}$; retain between thresholds. Exit on convergence $\mathbb{E}d \le \delta_{\text{exit}}$, $p \le p_{\text{off}}$, released adverse evidence $y < y_{\min}$ or $c_j < \underline{c}_j$, or fixed T^{th}; freeze thresholds and distinguish reasons. Waiting forbids new thesis risk. Closed † cancels entries and flattens thesis exposure, retaining delivery hedges; it is absorbing until a new record. Select live legs by quotes, tenor, liquidity/spread caps and delivery terms; retain held legs for exit. Use $EC, GE^\top$. Recompare *shares, calls, spreads, hedges and zero*. Expiry must cover holding or have a roll/expiry branch. An active thesis may hold zero. Baseline disables this gate and sets $\kappa_d = 0$. **To O5–O6.**	Probability/mode/deadline scalars; indices `(m,)`, selected arrays. Contract expiry, planning horizon and thesis deadline are distinct.

SPY options: dynamic policy and a multi-week example

The action is an instrument/hedge/exercise decision conditioned on a thesis belief. Assignment and future price paths remain random.

Objects: symbol, type, dimensions	Financial meaning, estimator / equation, and downstream consumer	Python representation
O5. Baseline and longer control Account $s^{\text{acct}} = (q, h, B, \mathfrak{p})$, $q \in \mathbb{Z}^m, h \in l_h\mathbb{Z}, B \in \mathbb{R}$; $\mathfrak{p}$: pending claims. $a = (\Delta q, \Delta h, e^+)$: bounded integer/lot trades and owned exercise. $s = (b, z, T^{\text{th}} - t, s^{\text{acct}}, \text{pending orders})$.	Baseline $H = 1$: $a^* = \arg\max_{a \in \mathcal{A}(s)} P^{-1} \sum_p U(W_{k+1}^{\text{liq},(p)}(a))$, $U(w) = w - \lambda_U\sqrt{a_U^2 + w^2}$, $a_U > 0, 0 < \lambda_U < 1$. The ledger below generates each wealth sample. For $H > 1$, $Q_k^h(s,a) = \mathbb{E}_k^P[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{h-1}(s')]$, $V_k^h(s) = \max_{a \in \mathcal{A}(s)}[Q_k^h(s,a) - h_0 \mathbf{1}_{a_{\text{trade}} \neq 0}]$. At the deadline set $V^0 = U(W^{\text{liq}})$; earlier boundaries use this liquidation approximation or an estimated continuation value. One-period baseline has $\eta_T = h_0 = 0$. Expectations also average assignment/fills. Stochastic MPC executes the first action of a nonanticipative tree; full Bellman control plans to the deadline. Future actions see observations/posteriors, not latent states or later shocks. **To O6–O7.**	Finite action enumeration or global optimizer; CVXPY arrays for convex subproblems. D_T is a fixed positive diagonal trade-unit scale; $\eta_T, h_0 \geq 0$ are noncash preferences.
O5a. Risk / no-action region $\mathcal{A}(s)$: feasible controls; $b_G \in \mathbb{R}_+^3$, $\bar{q} \in \mathbb{Z}_+^m$. $\Pi_p = W_p'^{\text{liq}} - R_f W$, $R_f = e^{\int r\,\mathrm{d}t}$; $\alpha \in (0,1)$.	At every node/interim event enforce $\lvert q\rvert \leq \bar{q}$, $0 \leq e^+ \leq (q + \Delta q)_+$, legal exercise dates, $\lvert Gq + (h + h^{\text{pend}}, 0, 0)^\top\rvert \leq b_G$, and available cash minus broker margin $\geq R_{\text{cash}}$, including joint-assignment stresses. One-step tail constraint: $\widehat{\text{ES}}_\alpha(-\Pi) = \min_\zeta\{\zeta + [P(1-\alpha)]^{-1}\sum_p[-\Pi_p - \zeta]_+\} \leq L_{\text{ES}}$. If hold is feasible, the full no-action region is $\max_{a \neq 0}[Q^H(s,a) - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}] \leq Q^H(s,0)$, also comparing owned exercise. Cash costs are already in wealth; turnover/switch penalties add inertia. Mandatory exercise/risk exits and O4 closure override waiting. Use fixed ties. **To O5 feasible actions.**	P&L samples `(P,)` per action; conditional mean/variance by sample moments; ES hinges; record value of holding and best alternative.
O6. Optimal execution is separate $y_e, v_e \in \mathbb{R}^{m+1}$: remaining parent trade and child rate; $T_e \ll \Delta_d$. $\Lambda_e \succ 0, \Omega_e \succeq 0 \in \mathbb{R}^{(m+1)\times(m+1)}$.	Given O5's trade target, minimize $\int_0^{T_e}[s_e^\top\lvert v_e\rvert + v_e^\top\Lambda_e v_e + \lambda_e y_e^\top\Omega_e y_e]\,\mathrm{d}u$, $\dot{y}_e = -v_e$, $y_e(0) = a_{\text{trade}}$, $y_e(T_e) = 0$, with lot/rate/legging/margin limits. Fit impact from fills and Ω_e from short-horizon quote moves. Replace O5's forecast cash costs by expected execution costs; its risk penalty is not cash. Immediate full fills remain baseline. Pending/unfilled legs return to strategic state; owned-exercise instructions bypass this trade scheduler. **To O7 fills/exercise.**	Execution optimizer/callable, arrays `(m+1,)`, order book and fill records. Allocation selects exposure; execution schedules it.
O7. Cash, delivery and assignment $K \geq 0$: actual trade cost; $e = e^+ - A \in \mathbb{Z}^m$. A_j: assigned shorts; $\beta_A, f_j \in \mathbb{R}^5$. $B^{\text{pend}}, h^{\text{pend}} \in \mathbb{R}$: unsettled cash/shares.	For actual fills: $B' = B - C^\top\Delta q - S\Delta h - K$, $q' = q + \Delta q, h' = h + \Delta h$. On exercise, $q'' = q' - e$, $\Delta h^{\text{pend}} = \sum m_j\chi_j e_j$, $\Delta B^{\text{pend}} = -\sum m_j K_j\chi_j e_j$. Transfer at delivery once. Pre-expiry baseline $A_j \sim \text{Binom}((-q_j')_+, \sigma(\beta_A^\top f_j))$, independent conditionally; $\sigma(x) = (1 + e^{-x})^{-1}$, $f = (1, \log(S/K), T - t, C - mg, D_{\text{next}}^{\text{div}})$. Fit bounded β_A by binomial likelihood. At expiry use the declared exercise/contrary-notice rule; remove unexercised legs. Notices are immediate; delivery follows its calendar. Cash accrues as $B \leftarrow R_\ell[B]_+ - R_b[-B]_+ - \text{Borrow} + h^{\text{ent}}D^{\text{div}}$. $W = B + B^{\text{pend}} + (h + h^{\text{pend}})S + q^\top C$. Liquidate at executable costs for W^{liq}; actual future marks replace predicted $\mathbb{Q}$ marks. Baseline $K = s^\top\lvert a_{\text{trade}}\rvert + a_{\text{trade}}^\top\Lambda a_{\text{trade}}$, $\Lambda \succeq 0$: observed spreads/fees and fitted impact, held fixed in forecasts. **To O5 and next posterior/account.**	Event ledger / pending-claim records, assignment samples `(P,H,m)`; cash/borrow/entitlement schedules as supplied functions. Only owned exercise is controlled.
O8. RL and validation $Q_\psi : \mathbb{R}^{d_s+d_a} \to \mathbb{R}$, $d_a = 2m + 1$; $\varphi_s(s) \in \mathbb{R}^{d_s}$: sufficient finite belief/account encoding.	Optional RL replaces O5's value solve: target $Y = -\eta_T\|D_T a_{\text{trade}}\|_1 - h_0\mathbf{1}_{a_{\text{trade}} \neq 0} + \max_{a' \in \mathcal{A}(s')} Q_{\bar{\psi}}(s', a')$, terminal maximum replaced by $U(W^{\text{liq}})$; loss $\mathbb{E}[(Q_\psi(s,a) - Y)^2]$. Useful for expensive repeated solves or learned action-dependent fills/information; unnecessary for tractable identified MPC. Require supported actions and validated simulator/offline data; preserve hard constraints. Seal chronological tests of path/catalyst calibration, $\mathbb{P}/\mathbb{Q}$ errors, assignment, costs, turnover and net benchmark gains. Blocks must preserve horizon overlap and longer dependence; adjust multiplicity. Gate solves only for valid/nonempty problems; otherwise funded exit or stop/default. **To control/update gate.**	Model callable / tensors; training tuples (s, a, r, s'). Audit posterior, gap, thesis mode, held-vs-switch value, instrument and exit reason. RL does not create the thesis forecast.

Worked conceptual example (hypothetical, normalized prices). Daily decisions; intraday observations; deadline day 20; a call expires day 40. Let filtered $u \sim \mathcal{N}(\log 110, 0.04^2)$, $\delta = 0$, $p_{\text{on}} = 0.90, p_{\text{off}} = 0.60$, $\delta_{\text{exit}} = 0.01$. **Entry:** at $S = 100$, $p \approx 0.99$; enter only if the call wins the $\mathbb{P}$-path/$\mathbb{Q}$-mark comparison after costs. **Noise:** an intraday dip with unchanged thesis evidence does not trigger a decision or invalidate it. **Week 2:** at $S = 106$, remaining gap $\log(110/106) = 0.037$, $p \approx 0.82$: retain the thesis. If the after-cost value gain from a smaller call spread (or shares) exceeds h_0, switch; otherwise hold. **Exit:** $S = 109$ gives gap $0.009 < \delta_{\text{exit}}$, so close on convergence. Alternatively a failed catalyst moving the anchor to 102 at $S = 106$ gives $p \approx 0.17$: invalidate. Day 20 closes any remaining thesis risk; zero exposure can be optimal earlier while the thesis remains active. **Composition:** $O \to (X, y) \to (b, \widehat{\theta}, d, z) \to \mathsf{P}_k^H \to \{c^{A,Q}(Z_l^{(p)}), W_l^{(p)}\} \to (Q^H, \mathcal{A}) \to a^{\text{str}} \to \text{fills} \to W' \to b'$.

$$a_k^* = F(O_{\leq k}, s_k^{\text{acct}}; b_k, \widehat{\theta}, \widehat{\beta}_A) = \mathcal{E}_{\text{O6}}\left[\text{Gate}_{\text{O4,O8}}\left(\arg\max_{a \in \mathcal{A}(s_k)}\left\{\mathbb{E}_{\substack{Z \sim \mathsf{P}_k^H,\ C_l = c^{A,Q}(Z_l) \\ s' = \text{Ledger}_{\text{O7}}(s_k, a)}}[-\eta_T\|D_T a_{\text{trade}}\|_1 + V_{k+1}^{H-1}(s')] - h_0\mathbf{1}_{a_{\text{trade}} \neq 0}\right\}\right)\right].$$